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Macroeconomic Regime Clustering & Tactical Asset Allocation

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By:

Yichen Gao (yg635)
Olivia Guo (qg77)
Fanyi Lyu (fl466)
Adarsh Pandey (ap2459)
Yixuan Wu (yw2694)
Yang Yue (yy923)

Faculty Advisor:

Prof. David Ruppert

Sponsor:

Vanguard Asset Management

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Faculty Advisor's Signature

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1 Project Overview

1.1 Introduction

Macroeconomic conditions drive broad shifts in asset returns, risk premia, and investor behavior. Static allocation frameworks cannot adapt to these changing environments, often taking on too much risk in downturns and too little in recoveries. This project investigates whether latent macro-financial regimes can provide a systematic foundation for adjusting portfolio risk in line with prevailing economic conditions.

1.2 Methodology

We estimate regimes using four complementary approaches—Time-Series K-Means, Bayesian Gaussian Mixture Models, Hidden Markov Models, and a Regularized Jump Model—each offering a distinct representation of macro states. These regime sequences are then combined with four allocation engines: Mean-Variance Optimization, Relative-Performance Tilting, Regime-Aware Benchmark Tilting, and a Regime-Adjusted Black-Litterman model. Together, these model combinations form a comprehensive framework for assessing how regime signals influence portfolio tilts and overall performance. We explore several regime identification techniques and evaluate their ability to cluster the macroeconomic environment using a broad set of economic and market features.

1.3 Executive Summary

Regime-aware asset allocation consistently improves portfolio behavior by becoming defensive in crisis regimes and more risk-seeking in expansions. This dynamic alignment enhances resilience and produces stronger risk-adjusted performance than static strategies. Across all experiments, four regime-allocation combinations emerge as the most practical and robust: TS-KMeans + Relative Performance, TS-KMeans + Regime-Aware Benchmark Tilt, HMM + Regime-Based MVO, and HMM + Regime-Adjusted Black-Litterman.

2 Literature Review

2.1 Time-Series K-Means

Time-Series K-means extends classical K-means to sequential data by replacing Euclidean distance with Dynamic Time Warping (DTW), enabling comparisons based on the *shape* of trajectories rather than strict pointwise alignment. This feature makes the method well-suited for macroeconomic datasets in which different indicators may move together but peak or trough at slightly different times.

Method Overview. The algorithm alternates between assigning each rolling window to the nearest centroid under DTW distance and updating centroids through DTW barycenters, yielding representative prototypes of typical macroeconomic dynamics. The approach remains nonparametric, computationally feasible, and interpretable.

Relevance for Regime Identification. Existing studies show that DTW-based clustering effectively recovers meaningful business-cycle states—such as expansions, slowdowns, crises, and recoveries—from high-dimensional macroeconomic time series. Its robustness to timing misalignment across indicators and its ability to capture joint dynamic patterns make TS K-means a natural component of our regime-identification toolkit, complementing more structured probabilistic models such as Gaussian HMMs and Regularized Jump Models.

2.2 Hidden Markov Models

Standard Hidden Markov Models (HMMs) have been widely used in finance for regime identification since Hamilton (1989). An HMM assumes a latent Markov chain $P(x_t = j \mid x_{t-1} = i) = A_{ij}$ and observations drawn from state-dependent emissions $y_t \sim b_{x_t}(y_t)$, allowing financial returns to exhibit evolving volatility, correlations, and distributional shape.

A key limitation is that all input features are treated as equally informative, with no mechanism to separate state-relevant variables from noise.

In “A Novel Dynamic Asset Allocation System Using Feature Saliency Hidden Markov Models for Smart Beta Investing” (Fons et al., 2019), this issue is addressed via the **Feature Saliency HMM (FSHMM)** of Adams et al. (2016). FSHMM introduces a relevance indicator z_l and saliency parameter $\rho_l = P(z_l = 1)$, modifying the emission likelihood to

$$p(y_{t,l} \mid x_t = i) = \rho_l, r(y_{t,l} \mid \mu_{il}, \sigma_{il}^2)(1 - \rho_l), q(y_{t,l} \mid \epsilon_l, \tau_l^2),$$

a mixture of regime-dependent and noise components.

Estimating ρ_l jointly with the HMM enables *simultaneous regime detection and feature selection*, improving robustness, mitigating overfitting, and allowing the model to scale to high-dimensional macro or factor panels.

Thus, FSHMM extends classical HMMs by learning which variables truly drive the latent regimes.

2.3 Gaussian Mixture Model

Botte and Bao (2021) propose a data-driven approach to identifying financial market regimes using Gaussian Mixture Models (GMMs). Instead of relying on expert-defined categories, they apply unsupervised learning to 17 factors from the Two Sigma Factor Lens (1970s–2020). The GMM represents factor returns as a mixture of Gaussian components:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (1)$$

with mixing weights π_k , means $\boldsymbol{\mu}_k$, and covariance matrices $\boldsymbol{\Sigma}_k$. Cross-validation selects four regimes: *Crisis* (17%), marked by negative equity and credit performance; *Steady State* (49%), exhibiting broad positive returns; *Inflation* (15%), driven by strong inflation hedges and USD weakness; and *Walking on Ice* (19%), characterized by elevated volatility and fragility.

These regimes correspond to major historical episodes such as the 1970s inflation era, the 1987 crash, the 2008 Global Financial Crisis, and the 2020 COVID shock. The model also highlights the dominance of Steady State conditions during the 2010s under accommodative monetary policy.

For portfolio use, the authors emphasize two applications. First, *risk management and scenario analysis*: sampling from regime-specific distributions enables stress testing using distinct return, volatility, and correlation structures. Second, *asset allocation*: while regime probabilities can inform tactical views, the authors stress that GMMs are descriptive rather than predictive, reinforcing the need for diversification across regimes. A Bayesian GMM variant further allows parameter uncertainty quantification and incorporation of prior beliefs.

2.4 Jump Models

Jump Models offer a nonparametric alternative to classical HMMs by modeling regime changes through explicit penalties on switching rather than assuming parametric transition dynamics [4]. Subsequent variants—including Sparse and Regularized Jump Models—extend this idea to high-dimensional financial settings by introducing feature selection and stability-oriented regularization. These developments have made Jump Models a practical and interpretable tool for identifying economic regimes and supporting regime-aware portfolio decisions.

Choice of Model. In high-dimensional macro-financial datasets, regime methods must control overfitting and avoid unstable variable selection. Standard Jump Models treat all features equally and may be sensitive to noise, while sparse variants can produce inconsistent sparsity patterns. The Regularized Jump Model (RJM) addresses these limitations by imposing structured penalties that deliver more stable, interpretable feature importance and more reliable regime boundaries [3]. This balance of flexibility and robustness makes RJM the most appropriate Jump Model variant for our empirical setting.

3 Datasets

Our dataset combines macroeconomic and market features primarily sourced from the FRED_MD database and additional Bloomberg indicators.

We used quantitative techniques such as correlation pruning and principal component analysis (PCA) to reduce the full database from over 140 features to 41 core variables. The dataset was first partitioned into structured macroeconomic groups: Output & Income, Labor Market, Consumption & Orders, Orders & Inventories, Money & Credit, Interest & Exchange Rates, Prices, and Stock Market—to ensure that feature selection preserved economic interpretability rather than collapsing all variables into a single undifferentiated pool. Within each subgroup, we performed correlation pruning by discarding features exhibiting more than 80% pairwise correlation, preventing redundant signals from dominating the clustering algorithms. After pruning, we applied PCA separately within each group and selected either the top five variables with the highest absolute loadings on the first principal component or, when fewer variables remained, retained all surviving indicators. This procedure captures the dominant variance-driving signals within each economic block while avoiding overrepresentation of any single category. The resulting feature set is both dimensionally efficient and economically diverse, providing a balanced and information-rich input space for downstream regime-identification models.

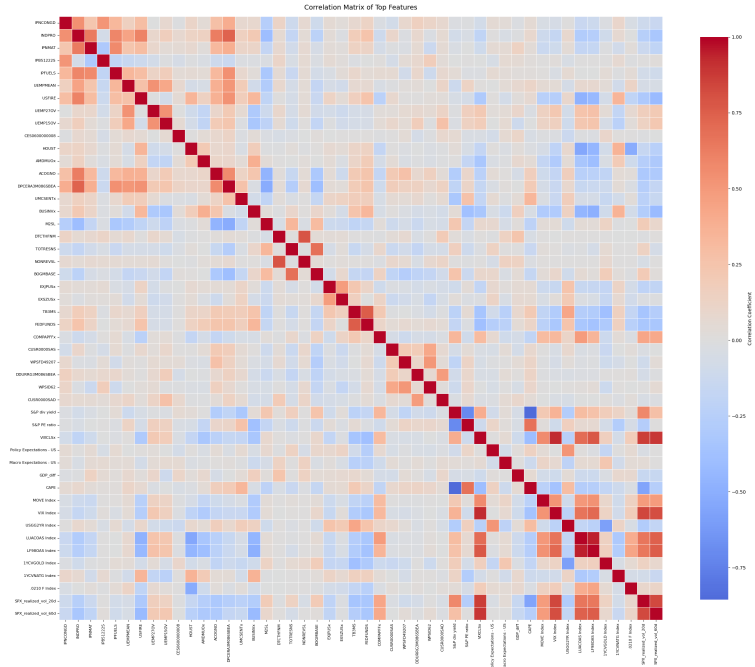


Figure 1: Heatmap of selected features

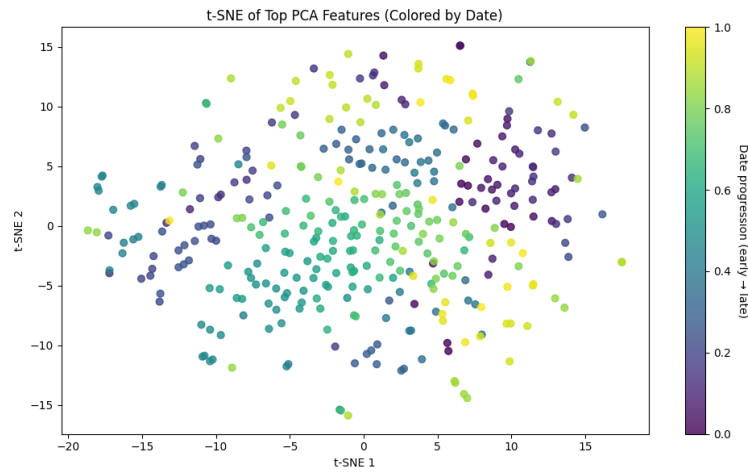


Figure 2: T-Sne plot of selected features

We observe from the heatmap that the pruned dataset contains fewer highly correlated variables, indicating that redundant information has been effectively removed. The t-SNE projection further shows that the clusters preserve a clear temporal structure even after feature reduction. This confirms that the essential dynamics of the dataset are retained, and the reduced feature set still captures the underlying macroeconomic evolution.

The final set of 41 selected features, obtained after correlation pruning and subgroup-level PCA-based filtering, is summarized in Table below:

Group	Features	Count	Descriptions
Output & Income	IPNCONGD, INDPRO, IPNMAT, IPB51222S, IPFUELS	5	IP Nondurable Consumer Goods; Industrial Production Index; IP Nondurable Materials; IP Residential Utilities; IP Fuels
Labor Market	UEMPMEAN, USFIRE, UEMP27OV, UEMP15OV, CES0600000008	5	Average Unemployment Duration; Employment in Financial Activities; Unemployed 27+ Weeks; Unemployed 15+ Weeks; Average Hourly Earnings (Goods-Producing)
Consumption & Orders	HOUST, AMDMUOx, DPCERA3M086SBEA, UMCSENTx, BUSINVx	5	Housing Starts; Unfilled Durable Orders; Real Personal Consumption Expenditures; Consumer Sentiment Index; Total Business Inventories
Money & Credit	M2SL, DTCTHFNM, TOTRESNS, NONREVSL, BOGMBASE	5	M2 Money Stock; Total Consumer Loans and Leases; Total Reserves; Non-revolving Credit; Monetary Base
Interest Rate & FX Rates	EXJPUSx, EXSZUSx, TB3MS, FEDFUNDS, COMPAPFFx	5	Japan/US FX; Switzerland/US FX; 3-Month Treasury Bill; Federal Funds Rate; Commercial Paper – Fed Funds Spread
Prices	CUSR0000SAS, WPSFD49207, DDURRG3M086SBEA, WPSID62, CUSR0000SAD	5	CPI Services; PPI Finished Consumer Goods; PCE Durables Deflator; PPI Durables; CPI Durables
Stock Market	S&P dividend yield, S&P PE ratio	2	S&P Composite Dividend Yield; S&P Price-Earnings Ratio
Other / Derived Indicators	GDP_diff, CAPE, MOVE Index, VIX Index, USGG2YR Index, LUACOAS Index, .0210F Index, SPX_realized_vol_20d, SPX_realized_vol_60d	9	Derived or market indicators: GDP gap, valuation (CAPE), bond/volatility measures, yield spreads, and realized volatility metrics

Table 1: Overview of macroeconomic and market features used in the dataset.

4 Methodology and Models

We implement four different regime identification techniques and compare their behavior across the macroeconomic cycle.

4.1 Time-Series K-means

We apply a temporal K-means framework to cluster economic conditions based on the joint trajectory of macroeconomic indicators.

4.1.1 Time-Series K-Means

In this section we describe our time-series clustering model for identifying macroeconomic regimes using Dynamic Time Warping (DTW) K-means. The model clusters trajectories of macro variables over rolling windows, so that regimes are defined by the joint evolution of macro and market conditions rather than by point-in-time levels alone. We present both a static specification with a fixed train-test split and a semiannual expanding-window specification for improved real-time adaptiveness.

4.1.2 Data Preparation and Expanding Standardization

Let $x_{t,i}$ denote the raw value of feature i at month t . As in the other regime models, we standardize each feature using an expanding window to avoid look-ahead bias. For t greater than a burn-in of 24 months, we compute

$$z_{t,i} = \frac{x_{t,i} - \mu_{t-1,i}}{\sigma_{t-1,i}}, \quad (2)$$

where $\mu_{t-1,i}$ and $\sigma_{t-1,i}$ are the sample mean and standard deviation of feature i using only observations up to month $t - 1$. This ensures that all inputs to the clustering algorithm are strictly backward looking and therefore feasible in real time.

4.1.3 Rolling Window Construction

To capture short-run macro dynamics, we transform the standardized monthly series $\{z_t\}$ into overlapping fixed-width windows. For a window length of $W = 12$ months, each time-series observation is defined as

$$W_t = (z_{t-W+1}, z_{t-W+2}, \dots, z_t), \quad t = t_{\min} + W - 1, \dots, t_{\max}, \quad (3)$$

where each z_s is the vector of standardized features at time s . The end-of-window date t serves as the timestamp and will receive a single regime label. This construction allows the clustering to depend on both levels and recent changes in the macro variables.

4.1.4 Dynamic Time Warping and TS K-Means Objective

Given two sequences $A = (a_1, \dots, a_T)$ and $B = (b_1, \dots, b_T)$ in $\mathbb{R}^d$, the DTW distance is defined as

$$\text{DTW}(A, B) = \min_{\pi} \sum_{(i,j) \in \pi} \|a_i - b_j\|_2^2, \quad (4)$$

where π ranges over all monotone warping paths that align indices of A and B . DTW permits local time shifts and therefore measures similarity in the *shape* of trajectories even if peaks or troughs occur at slightly different dates, a desirable property for macroeconomic cycles.

Time-Series K-means replaces the usual Euclidean distance with DTW in the K-means objective. Let $\{W_t\}$ denote the set of rolling windows and let $d(\cdot, \cdot)$ be the DTW distance. For a chosen number of clusters K , TS K-Means solves

$$\min_{\{C_k\}, \{\mu_k\}} \sum_{k=1}^K \sum_{W_t \in C_k} d(W_t, \mu_k)^2, \quad (5)$$

where C_k is the set of windows assigned to cluster k and μ_k is the DTW barycenter (cluster centroid trajectory) of that cluster.

4.1.5 Algorithm: Time-Series K-Means with DTW

We implement TS K-Means using the standard alternating minimization procedure:

Algorithm 1 Time-Series K-Means with DTW

- 1: **Input:** Rolling windows $\{W_t\}$; number of clusters K .
- 2: **Initialize:** Centroids $\{\mu_k^{(0)}\}_{k=1}^K$ by sampling K windows at random.
- 3: **for** $\ell = 0, 1, 2, \dots$ until convergence **do**
- 4: **Assignment step:** For each window W_t , assign it to cluster

$$c_t^{(\ell)} = \arg \min_{k \in \{1, \dots, K\}} d(W_t, \mu_k^{(\ell)}),$$

where $d(\cdot, \cdot)$ denotes the DTW distance.

- 5: **Update step:** For each $k = 1, \dots, K$, let

$$C_k^{(\ell)} = \{t : c_t^{(\ell)} = k\}$$

be the set of windows assigned to cluster k . Update the centroid as

$$\mu_k^{(\ell+1)} = \text{DTWBarycenter}(\{W_t : t \in C_k^{(\ell)}\}).$$

- 6: **end for**
 - 7: **Output:** Cluster assignments $\{c_t\}$ and centroids $\{\mu_k\}$. =0
-

4.1.6 Choosing the Number of Clusters

We evaluate the within-cluster DTW dispersion for $K \in \{2, \dots, 7\}$ and select K using the elbow method. Figure 3 plots the mean within-cluster DTW radius as a function of K . The curve exhibits a clear elbow at $K = 5$, with large improvements up to $K = 5$ and only marginal gains thereafter, so we set $K = 5$ for all subsequent analyses.

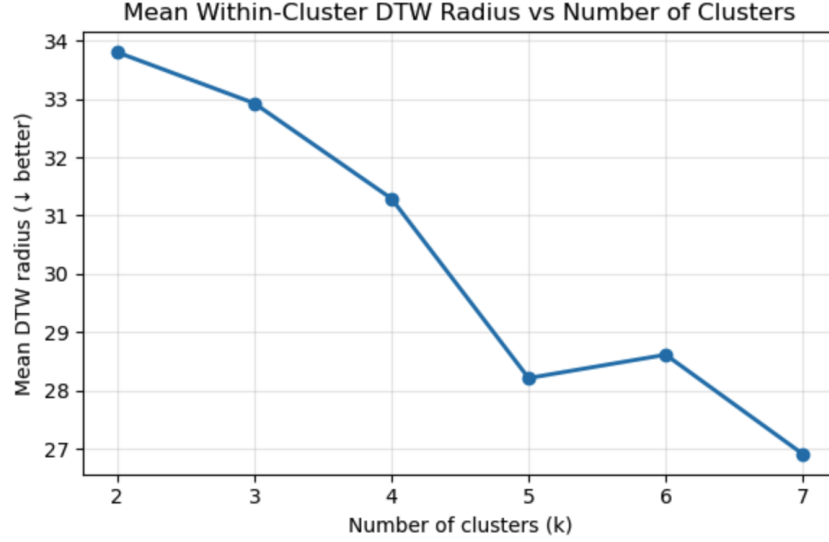


Figure 3: Mean within-cluster DTW radius as a function of the number of clusters K . An elbow at $K = 5$ motivates the choice of five regimes.

4.1.7 Static TS K-Means: Train–Test Setup

For the static specification, we estimate TS K-Means using all windows that end on or before December 2019. Windows from January 2020 onward are treated as out-of-sample and are assigned regime labels using the fitted centroids. This mirrors the train–test split used in the other regime models and preserves strict information separation.

Figure 4 shows the resulting regime timeline from 1992 to 2025. The model identifies persistent stretches of expansionary regimes, crisis episodes around 2000, 2008, and 2020, and subsequent recovery phases.

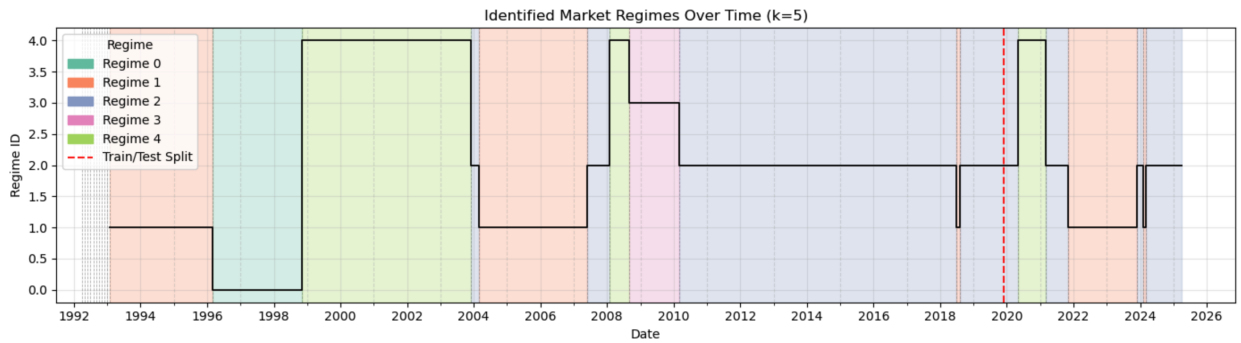


Figure 4: Static TS K-Means regime timeline with $K = 5$. The dashed red line marks the 2019–2020 train–test split.

4.1.8 Static TS K-Means: Economic Interpretation

To interpret the regimes economically, we compute within-regime averages of key macro and financial indicators. Table 2 summarizes the mean and standard deviation of selected features (industrial production growth, inflation, interest rates, financial conditions, volatility, valuation, leverage, unemployment and the S&P 500 level) for each regime. Table 3 provides compact labels and qualitative descriptions.

	Industrial Production (YoY) (x100)	Core CPI (MoM) (x1000)	2Y Treasury Yield Change	US Financial Conditions	VIX Volatility	CAPE Valuation	Margin Debt Growth	Unemployment Rate	SPX Index
Regime									
0	0.51 $\pm$ 0.51	-0.02 $\pm$ 0.87	-0.04 $\pm$ 0.25	0.68 $\pm$ 0.22	22.43 $\pm$ 6.72	0.24 $\pm$ 1.08	6.67 $\pm$ 0.34	4.98 $\pm$ 0.39	860.48 $\pm$ 165.49
1	0.29 $\pm$ 0.51	0.01 $\pm$ 1.51	0.05 $\pm$ 0.27	0.82 $\pm$ 0.11	13.24 $\pm$ 1.81	0.07 $\pm$ 0.5	6.66 $\pm$ 2.03	5.58 $\pm$ 0.82	895.64 $\pm$ 439.45
2	0.11 $\pm$ 0.47	0.03 $\pm$ 0.91	-0.02 $\pm$ 0.17	1.4 $\pm$ 0.34	17.31 $\pm$ 5.38	0.07 $\pm$ 0.72	10.15 $\pm$ 1.11	6.1 $\pm$ 1.99	1921.13 $\pm$ 589.3
3	-0.5 $\pm$ 1.58	-0.08 $\pm$ 1.17	-0.09 $\pm$ 0.25	3.57 $\pm$ 1.6	34.49 $\pm$ 11.94	-0.08 $\pm$ 1.27	7.03 $\pm$ 0.4	8.56 $\pm$ 1.42	981.86 $\pm$ 138.57
4	0.03 $\pm$ 0.51	-0.03 $\pm$ 1.77	-0.03 $\pm$ 0.31	1.65 $\pm$ 0.5	24.29 $\pm$ 4.51	-0.15 $\pm$ 1.35	6.96 $\pm$ 0.8	4.93 $\pm$ 0.82	1197.02 $\pm$ 193.22

Table 2: Summary statistics of key macro and financial features by TS K-Means regime (static model). Values are reported as mean $\pm$ standard deviation.

Regime ID	Label	Key Features
0	Mid-Cycle Stable Growth	Moderate growth; easy financial conditions; stable valuations; moderate volatility.
1	Healthy Expansion	Solid growth; low volatility; strong financial and liquidity backdrop.
2	Late-Cycle Softening	Slowing growth; neutral conditions; rising uncertainty.
3	Crisis / Shock	Negative growth; tight financial conditions; high volatility.
4	Early-Cycle Recovery	Improving growth; falling yields; easing volatility; strong liquidity support.

Table 3: Economic labels and qualitative descriptions for the five TS K-Means regimes.

4.1.9 Static TS K-Means: Forward-Return Separability

We next examine whether regimes are associated with distinct forward returns. For each regime and each asset, we compute the mean and standard deviation of 6-month forward returns over the train and test samples, and assess cross-regime separation using ANOVA.

Tables 4 and 5 display the forward 6-month absolute returns for major asset classes (equities, aggregate bonds, gold and the dollar) in the train and test samples, respectively.

Forward 6M Asset Returns by Market Regime (Train Sample, with ANOVA significance)						
	Regime	Regime_Label	Russell 1000 (6M fwd)	US Agg Bond (6M fwd)	Gold (6M fwd)	Dollar (6M fwd)
0	0.000000	Mid-Cycle Stable Growth	0.128 ± 0.090	0.043 ± 0.020	-0.054 ± 0.043	0.032 ± 0.038
1	1.000000	Healthy Expansion	0.056 ± 0.057	0.024 ± 0.032	0.057 ± 0.090	0.006 ± 0.039
2	2.000000	Late-Cycle Softening	0.045 ± 0.079	0.020 ± 0.025	0.033 ± 0.114	0.010 ± 0.040
3	3.000000	Crisis / Shock	0.087 ± 0.166	0.045 ± 0.016	0.118 ± 0.066	-0.012 ± 0.054
4	4.000000	Early-Cycle Recovery	-0.030 ± 0.145	0.031 ± 0.029	0.028 ± 0.086	0.012 ± 0.049
5		ANOVA p-value	0.0000	0.0000	0.0000	0.0062

Table 4: Forward 6-month asset returns by TS K-Means regime (train sample, static model). The bottom row reports ANOVA p-values for cross-regime equality of means.

Forward 6M Asset Returns by Market Regime (Test Sample, with ANOVA significance)						
	Regime	Regime_Label	Russell 1000 (6M fwd)	US Agg Bond (6M fwd)	Gold (6M fwd)	Dollar (6M fwd)
0	1.000000	Healthy Expansion	0.036 ± 0.116	-0.011 ± 0.052	0.049 ± 0.101	0.012 ± 0.039
1	2.000000	Late-Cycle Softening	0.076 ± 0.087	0.011 ± 0.035	0.133 ± 0.083	0.004 ± 0.040
2	4.000000	Early-Cycle Recovery	0.185 ± 0.051	-0.006 ± 0.016	-0.021 ± 0.069	-0.030 ± 0.029
3		ANOVA p-value	0.0005	0.1428	0.0000	0.0144

Table 5: Forward 6-month asset returns by TS K-Means regime (test sample, static model).

The train sample shows strong regime separation (p-values near zero across assets). Equities perform best in the mid-cycle stable growth and crisis/rebound regimes, while bonds and gold deliver their strongest returns in crisis and late-cycle softening states. The dollar tends to strengthen in expansions and weaken in downturns. In the post-2020 test sample, statistical significance weakens but remains economically meaningful, particularly for equities and gold.

We also examine relative performance across equity and credit styles. Tables 6 and 7 summarize 6-month forward spreads for Growth–Value, High Yield–Investment Grade, long–short Treasury, and EM–DM equities.

Forward 6M Relative Performance by Market Regime (Train Sample)						
	Regime	Regime_Label	Growth - Value (6M fwd)	HY - IG (6M fwd)	Long - Short Treasury (6M fwd)	EM - DM Equities (6M fwd)
0	0.000000	Mid-Cycle Stable Growth	0.037 ± 0.06	-0.001 ± 0.041	0.027 ± 0.039	-0.098 ± 0.145
1	1.000000	Healthy Expansion	0.001 ± 0.045	0.017 ± 0.026	0.011 ± 0.051	0.051 ± 0.127
2	2.000000	Late-Cycle Softening	0.03 ± 0.053	0.005 ± 0.043	0.032 ± 0.078	-0.005 ± 0.061
3	3.000000	Crisis / Shock	0.026 ± 0.063	0.088 ± 0.1	0.01 ± 0.065	0.129 ± 0.089
4	4.000000	Early-Cycle Recovery	-0.024 ± 0.134	-0.02 ± 0.081	0.012 ± 0.046	0.038 ± 0.112
5		ANOVA p-value	0.0000	0.0000	0.0878	0.0000

Table 6: Forward 6-month relative performance by TS K-Means regime (train sample): Growth–Value, HY–IG, long–short Treasury and EM–DM equity spreads.

Forward 6M Relative Performance by Market Regime (Test Sample)						
	Regime	Regime_Label	Growth - Value (6M fwd)	HY - IG (6M fwd)	Long - Short Treasury (6M fwd)	EM - DM Equities (6M fwd)
0	1.000000	Healthy Expansion	0.033 $\pm$ 0.11	0.023 $\pm$ 0.02	-0.06 $\pm$ 0.08	-0.04 $\pm$ 0.039
1	2.000000	Late-Cycle Softening	0.078 $\pm$ 0.107	0.017 $\pm$ 0.022	-0.008 $\pm$ 0.059	-0.003 $\pm$ 0.061
2	4.000000	Early-Cycle Recovery	-0.033 $\pm$ 0.09	0.056 $\pm$ 0.026	-0.069 $\pm$ 0.063	0.003 $\pm$ 0.087
3		ANOVA p-value	0.0205	0.0000	0.0108	0.0380

Table 7: Forward 6-month relative performance by TS K-Means regime (test sample).

The patterns broadly align with intuition: Growth outperforms Value in expansionary regimes, while Value leads in early-cycle recovery; HY spreads widen in crises and compress in recoveries; and long duration tends to outperform in late-cycle environments.

4.1.10 Expanding-Window TS K-Means

The static clustering specification estimates a single TS K-Means model on a fixed historical sample. To better capture the evolving macroeconomic structure and mimic real-time learning, we implement an expanding-window TS K-Means procedure that re-estimates the clustering model every six months using all data available up to that point.

Before applying this expanding-window model to guide portfolio allocation, it is essential to assess whether the regime identification is temporally stable. Following the approach described in the accompanying analysis :contentReference[oaicite:1]index=1, we evaluate three representative model snapshots trained using data up to 2010, 2015, and 2020. Figures 5–7 display the resulting regime assignments.

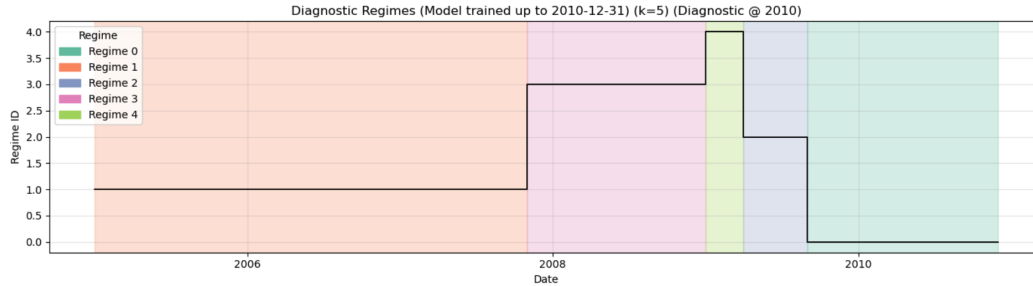


Figure 5: Diagnostic regimes from an expanding-window TS K-Means model trained through 2010 (evaluated at 2010).

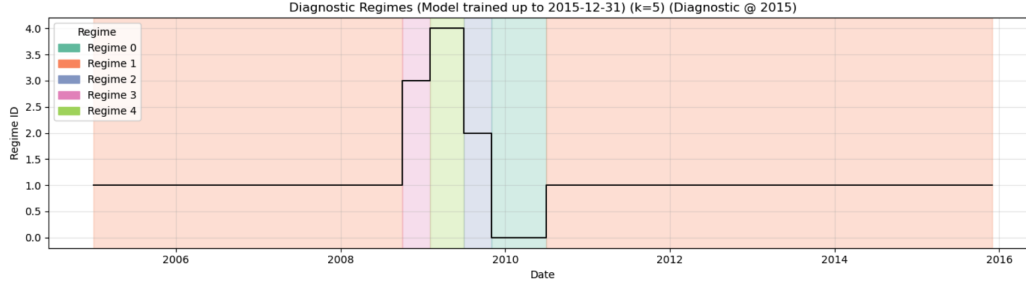


Figure 6: Diagnostic regimes from an expanding-window TS K-Means model trained through 2015 (evaluated at 2015).

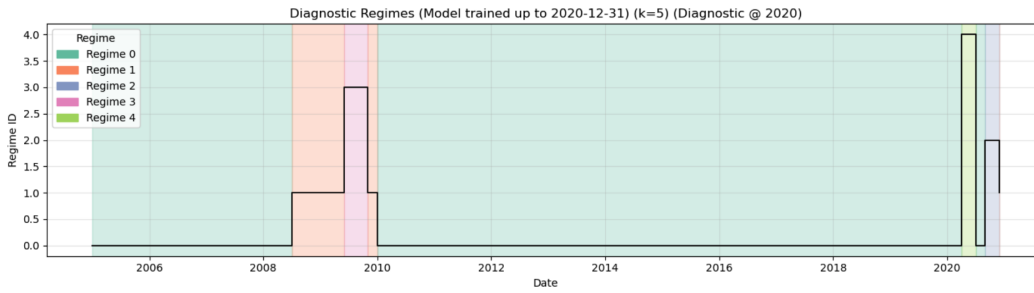


Figure 7: Diagnostic regimes from an expanding-window TS K-Means model trained through 2020 (evaluated at 2020).

Table 8: Regime Mapping Between 2020, 2015, and 2010

2020 Regime	Matching 2015 Regimes	Matching 2010 Regimes
0	1, 0	1, 0
1	3, 4, 0	3, 4, 0
2	<i>None</i>	<i>None</i>
3	2	2
4	<i>None</i>	<i>None</i>

Across the 2010, 2015, and 2020 snapshots, the TS K-Means model produces regime assignments that remain highly consistent over time. The close correspondence of regimes—visible in both the mapping table and the feature comparison—shows that the clustering algorithm is capturing persistent macro-financial structures rather than noise.

In particular, 2020 Regime 3 closely aligns with 2015 Regime 2, and their macro profiles remain nearly unchanged even after incorporating major events such as the 2018–2019 tightening cycle and the onset of COVID-19. Table 9 illustrates this stability across key indicators.

The strong temporal robustness of the expanding-window TS K-Means model implies that regime definitions remain stable as new data arrive. This consistency is essential for systematic asset allocation, ensuring that regime signals remain reliable and economically interpretable over time.

Table 9: Comparison of 2015 Regime 2 and 2020 Regime 3 Across Key Macro-Financial Indicators

Feature	2015 – Regime 2	2020 – Regime 3
Industrial Production (YoY) $\times 100$	0.84 ± 0.43	0.61 ± 0.64
Core CPI (MoM) $\times 1000$	0.37 ± 1.31	0.46 ± 1.15
2Y Treasury Yield Change $\times 1$	0.05 ± 0.06	0.01 ± 0.12
US Financial Conditions $\times 1$	2.26 ± 0.18	2.42 ± 0.39
VIX Volatility $\times 1$	27.06 ± 2.43	26.92 ± 2.13
CAPE Valuation $\times 1$	0.74 ± 0.47	0.67 ± 0.44
Margin Debt Growth $\times 1$	6.72 ± 0.07	6.73 ± 0.09
Unemployment Rate $\times 1$	9.6 ± 0.14	6.68 ± 0.51
SPX Index Level $\times 1$	996.13 ± 58.57	983.65 ± 126.11

4.1.11 Discussion

Taken together, the static and expanding-window TS K-Means results suggest that clustering macro trajectories with DTW provides a flexible and interpretable regime framework. The model

- avoids parametric distributional assumptions,
- is robust to local timing differences in macro indicators,
- yields regimes with clear macroeconomic meanings, and
- produces forward-return profiles that are both statistically and economically distinct across regimes.

These properties make TS K-Means an effective model to identify market regimes.

4.2 Bayesian Gaussian Mixture Models

BGMM allows for soft probabilistic assignments of regime membership, enabling smoother transitions across evolving macroeconomic states.

4.2.1 Data Preparation

Monthly Alignment and Filtering Data are resampled to month-end using last-available-value, forward-filled up to 12 months. Features with $< 50\%$ completeness and periods with $< 80\%$ completeness are dropped.

Leak-Safe Robust Scaling Features are standardized using 24-month rolling median $\tilde{\mu}_{jt}$ and MAD $\tilde{\sigma}_{jt}$:

$$\bar{z}_{jt} = \text{clip} \left(\frac{x_{jt} - \tilde{\mu}_{j,t-1}}{1.4826 \cdot \text{MAD}_{s=t-23}^{t-1}(x_{js}) + 10^{-8}}, -4, +4 \right) \quad (6)$$

The one-month lag ensures no look-ahead bias.

Whitening Transformation To improve BGMM stability and convergence, features are whitened using Ledoit-Wolf shrinkage covariance estimation.

Train-Test Split:

1. Estimate covariance $\hat{\Sigma}_{\text{train}}$ using Ledoit-Wolf on centered training data $\mathbf{Z}_{\text{train}} - \bar{\mathbf{Z}}_{\text{train}}$
2. Compute eigendecomposition: $\hat{\Sigma}_{\text{train}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$
3. Construct whitening matrix:

$$\mathbf{W}_{\text{train}}^{-1/2} = \mathbf{V}\mathbf{\Lambda}^{-1/2}\mathbf{V}^\top \quad (7)$$

4. Transform both train and test using training-only statistics:

$$\mathbf{X}_{\text{train}} = (\mathbf{Z}_{\text{train}} - \bar{\mathbf{Z}}_{\text{train}})\mathbf{W}_{\text{train}}^{-1/2} \quad (8)$$

$$\mathbf{X}_{\text{test}} = (\mathbf{Z}_{\text{test}} - \bar{\mathbf{Z}}_{\text{train}})\mathbf{W}_{\text{train}}^{-1/2} \quad (9)$$

This ensures no look-ahead bias—test data is transformed using only training period parameters.

4.2.2 Model Construction

The BGMM models the distribution of whitened features $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_T\}$ as a mixture of K Gaussian components:

$$p(\mathbf{x}_t | \boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_t | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (10)$$

where:

- π_k are mixture weights with $\sum_{k=1}^K \pi_k = 1$
- $\boldsymbol{\mu}_k \in \mathbb{R}^N$ are component means
- $\boldsymbol{\Sigma}_k \in \mathbb{R}^{N \times N}$ are component covariances

Dirichlet Process Prior on Mixture Weights A Dirichlet process (DP) prior is placed on the mixture weights:

$$\boldsymbol{\pi} \sim \text{DirichletProcess}(\alpha) \quad (11)$$

with concentration parameter $\alpha = 2.0$. This prior allows the model to automatically determine the effective number of components, shrinking weights of unnecessary components toward zero.

Gaussian Prior on Component Means

$$\boldsymbol{\mu}_k \sim \mathcal{N}(\mathbf{0}, \beta^{-1}\mathbf{I}) \quad (12)$$

where $\beta = 0.05$ is the mean precision prior, penalizing components with extreme means.

Tied Covariance Structure All components share a common covariance matrix:

$$\Sigma_k = \Sigma \quad \forall k \quad (13)$$

with regularization:

$$\Sigma_{\text{reg}} = \Sigma + \lambda \mathbf{I}, \quad \lambda = 5 \times 10^{-4} \quad (14)$$

This tied structure reduces model complexity and improves stability, particularly important given the whitening preprocessing.

4.2.3 Model Fitting via Variational Inference

The posterior distribution is approximated using variational Bayesian inference with the following hyperparameters:

Parameter	Value	Description
K_{max}	5	Maximum number of components
α	2.0	Weight concentration (Dirichlet Process)
β	0.05	Mean precision prior
λ	5×10^{-4}	Covariance regularization
Iterations	2000	Maximum EM iterations
Init	k-means	Initialization method

Table 10: BGMM hyperparameters

The model is fit on training data using the Expectation-Maximization (EM) algorithm, which alternates between:

- **E-step:** Compute responsibilities

$$\gamma_{tk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k, \Sigma)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_j, \Sigma)} \quad (15)$$

- **M-step:** Update parameters using posterior distributions incorporating priors

4.2.4 Post-Processing: Temporal Persistence

Hysteresis Smoothing First, we smooth regime probabilities using a 3-month rolling average:

$$\bar{\gamma}_{tk} = \frac{1}{3} \sum_{s=t-2}^t \gamma_{sk} \quad (16)$$

Regime switches occur only when there is strong evidence—the new regime must exceed 60% probability *and* the current regime must fall below 40%:

$$r_t = \begin{cases} k_{\text{new}} & \text{if } \bar{\gamma}_{t,k_{\text{new}}} > 0.60 \text{ and } \bar{\gamma}_{t,r_{t-1}} < 0.40 \\ r_{t-1} & \text{otherwise (maintain current regime)} \end{cases} \quad (17)$$

This hysteresis band (40%-60%) prevents rapid oscillations between regimes.

Minimum Duration Rule Any regime persisting fewer than 3 months is eliminated by merging into the adjacent regime with higher average probability during that period. This enforces economically meaningful persistence (regimes lasting 3-12 months).

4.2.5 Regime ID Handling Methodology

Market conditions in the test period may differ fundamentally from training. We allow the model to identify and create new regimes for these novel conditions.

Detecting Novel Observations For each month in the test period, we check how well the training regimes explain the data. If the highest probability across all training regimes is below 60%, we flag that month as "novel":

$$\text{Novel if: } \max_k \gamma_{tk}^{\text{train}} < 0.60 \quad (18)$$

Creating New Regimes We collect all novel months and fit a separate BGMM on just those observations. This identifies up to 1 new regime type that didn't exist during training (e.g., the COVID-19 crash period).

Combined Regime Assignment The final regime labels work as follows:

- **Training period:** Only training regimes are used
- **Normal test months:** Use training regimes (when maximum probability $\geq 60\%$)
- **Novel test months:** Use the new regime(s) fitted on novel data

This ensures training and normal test periods use consistent regime definitions, while extreme market conditions (like COVID) can be identified as distinct regimes.

To avoid too many regimes in the test period, we apply two rules:

1. **Remove rare regimes:** Any regime appearing in fewer than 3 test months is merged into the most common regime
2. **Limit total regimes:** If more than 3 regimes exist in test, we merge the least frequent ones based on similarity in feature space

Fitting Novel Regime Components Let $\mathcal{I}_{\text{novel}} = \{t : p_{\max}^{\text{train}}(\mathbf{x}_t) < \theta_{\text{novel}}\}$ denote the set of novel observations. A separate BGMM is fit on $\{\mathbf{x}_t : t \in \mathcal{I}_{\text{novel}}\}$ with up to $K_{\text{new}} = 1$ component(s):

$$p(\mathbf{x}_t \mid t \in \mathcal{I}_{\text{novel}}) = \sum_{k=K_{\text{train}}+1}^{K_{\text{train}}+K_{\text{new}}} \pi_k^{\text{new}} \mathcal{N}(\mathbf{x}_t \mid \boldsymbol{\mu}_k^{\text{new}}, \boldsymbol{\Sigma}^{\text{new}}) \quad (19)$$

Extended Responsibility Matrix The final regime probabilities combine training and novel components:

$$\gamma_{tk} = \begin{cases} \gamma_{tk}^{\text{train}} & \text{if } t \in \text{TRAIN or } (t \in \text{TEST and } \max_j \gamma_{tj}^{\text{train}} \geq 0.60), k \leq K_{\text{train}} \\ \gamma_{tk}^{\text{novel}} & \text{if } t \in \mathcal{I}_{\text{novel}}, k > K_{\text{train}} \\ 0 & \text{otherwise} \end{cases} \quad (20)$$

This ensures that:

- Training periods use only training regimes
- Normal test periods (where training regimes explain data well) use training regimes
- Novel test periods (where $\max_k \gamma_{tk}^{\text{train}} < 0.60$) use newly identified regimes
- Training regimes never appear during novel periods, and novel regimes never appear during training

Hierarchical Merging to $K_{\text{max}}^{\text{test}}$ Regimes If more than $K_{\text{max}}^{\text{test}} = 3$ regimes remain after pruning, we reduce the number through hierarchical merging:

1. Compute regime centroids $\mu_k^{\text{centroid}} = \frac{1}{|\{t:r_t=k\}|} \sum_{t:r_t=k} \mathbf{x}_t$ in whitened space
2. Rank regimes by frequency: $n_k = |\{t \in \text{TEST} : r_t = k\}|$
3. Retain the $K_{\text{max}}^{\text{test}} = 3$ most frequent regimes
4. For each non-retained regime k , find nearest neighbor among retained regimes:

$$k^* = \arg \min_{j \in \text{retained}} \|\mu_k^{\text{centroid}} - \mu_j^{\text{centroid}}\|_2 \quad (21)$$

5. Reassign all observations from regime k to regime k^*

This ensures the final test period has at most 3 distinct regimes, making results interpretable while preserving the most important market distinctions.

4.2.6 Regime Labeling: Economic Interpretation

Our framework classifies market environments into six distinct economic regimes. Classification proceeds hierarchically, with priority given to extreme conditions before moderate states:

1. **Crisis:** Characterized by severe negative returns ($r < -0.7\sigma$) accompanied by elevated volatility ($v > 0.6\sigma$) or extreme stress ($s > 0.6\sigma$). These periods represent systemic market dislocations.
2. **Correction:** Moderate negative returns ($-1.0\sigma < r < 0.1\sigma$) with rising volatility ($0.2\sigma < v \leq 0.7\sigma$) but stress levels below panic thresholds ($s < 0.6\sigma$). These represent typical drawdown periods without systemic risk.
3. **Walking on Ice:** Exceptional positive returns ($r > 1.0\sigma$) combined with elevated volatility or stress ($v > 0.1\sigma$ or $s > 0.1\sigma$). These regimes capture periods of strong performance amid fragile market conditions.
4. **Steady State:** Sustainable growth ($r > 0.4\sigma$) with calm volatility ($v < 0$) and low stress ($s < 0$). This represents the ideal environment for risk-taking.

5. **Inflation:** Positive returns ($r > 0.2\sigma$) accompanied by elevated volatility or stress ($v > 0.3\sigma$ or $s > 0.3\sigma$). Captures periods of growth with persistent uncertainty, typically associated with inflationary pressures.
6. **Mid Expansion:** The neutral baseline, characterized by moderate performance across all dimensions ($-0.3\sigma \leq r \leq 0.3\sigma$, $-0.2\sigma \leq v \leq 0.2\sigma$, $-0.2\sigma \leq s \leq 0.2\sigma$). This also serves as the default category for regimes not matching other criteria.

Classification Algorithm For each regime k identified by the statistical model, we compute the following standardized metrics across all time periods t where regime k is active:

$$r_k = \text{z-score}(\mathbb{E}[\text{return}_t \mid \text{regime}_t = k]) \quad (22)$$

$$v_k = \text{z-score}(\text{Std}[\text{return}_t \mid \text{regime}_t = k]) \quad (23)$$

$$s_k = \text{z-score}\left(\frac{1}{3} \sum_{i \in \{\text{VIX, HY OAS, MOVE}\}} \mathbb{E}[i_t \mid \text{regime}_t = k]\right) \quad (24)$$

where all z-scores are computed *across regimes* (not across time). Specifically:

$$r_k = \frac{\mathbb{E}[\text{return}_t \mid r_t = k] - \mu_{\text{regime}}}{\sigma_{\text{regime}}} \quad (25)$$

$$\text{where } \mu_{\text{regime}} = \frac{1}{K} \sum_{j=1}^K \mathbb{E}[\text{return}_t \mid r_t = j] \quad (26)$$

$$\sigma_{\text{regime}} = \sqrt{\frac{1}{K-1} \sum_{j=1}^K (\mathbb{E}[\text{return}_t \mid r_t = j] - \mu_{\text{regime}})^2} \quad (27)$$

This normalizes regime characteristics relative to each other, making the taxonomy robust to structural changes in market volatility or interest rate levels.

The classification function $\mathcal{C} : \mathbb{R}^3 \rightarrow \{\text{Crisis, Correction}, \dots, \text{Mid Expansion}\}$ evaluates conditions in the following priority order: (1) Crisis, (2) Correction, (3) Walking on Ice, (4) Steady State, (5) Inflation, (6) Mid Expansion criteria. Regimes not satisfying any specific criteria default to Mid Expansion.

4.2.7 Results

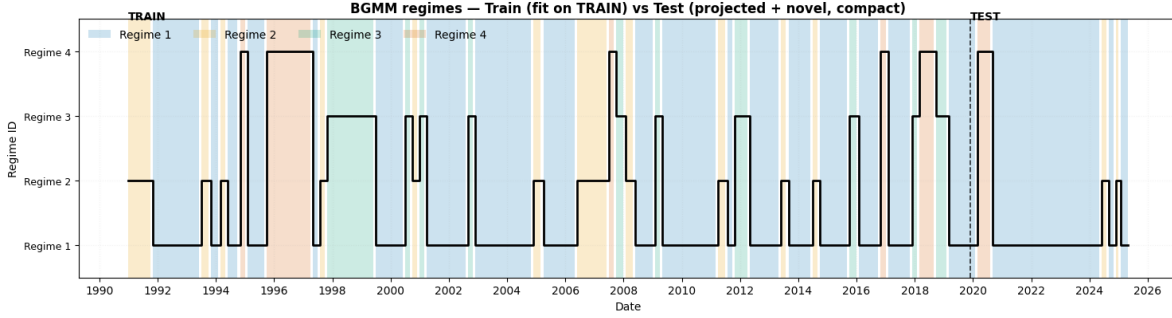


Figure 8: BGMM Regime Detection - Train/Test

Training Period (1990-2019): The model identifies four regimes during the 348-month training period, distributed as follows:

Table 11: Training Period Regime Distribution

Regime ID	Economic Label	Months	Frequency
Regime 1	Mid Expansion	205	58.91%
Regime 2	Steady State	54	15.52%
Regime 3	Steady State	54	15.52%
Regime 4	Crisis	35	10.06%

Test Period (2020-2025): The out-of-sample test period spans 65 months with markedly different regime composition:

Table 12: Test Period Regime Distribution

Regime ID	Economic Label	Months	Frequency
Regime 1	Mid Expansion	54	83.08%
Regime 2	Steady State	5	7.69%
Regime 4	Crisis	6	9.23%

Key Findings: **1. Crisis Detection:** The model successfully identifies the COVID-19 crash (6 months, 9.23%) using only pre-2020 training parameters, demonstrating that crisis characteristics learned from 1998, 2001, and 2008 generalize to unprecedented shocks.

2. Post-COVID Shift: Mid Expansion prevalence increases from 59% to 83% in the test period, indicating markets operate with moderate returns but elevated volatility—lacking the calm conditions required for Steady State classification.

3. Temporal Stability: Regime durations span 3-12 months with no rapid switching, confirming effective persistence smoothing. Transitions follow economically sensible patterns: sharp crisis entries followed by gradual recoveries.

4. Out-of-Sample Robustness: Test regimes remain interpretable despite using frozen training parameters, indicating the model learned generalizable features rather than overfitting to historical noise.

4.2.8 Forward-Return Separability

Forward 12M Asset Returns by Market Regime (Train Sample, with ANOVA significance)					
Regime	Regime_Label	Russell 1000	US Agg Bond	Gold	Dollar
0	Mid Expansion	0.091 ± 0.137	0.056 ± 0.043	0.088 ± 0.16	0.019 ± 0.06
1	Steady State	0.025 ± 0.163	0.067 ± 0.046	0.064 ± 0.156	0.03 ± 0.077
2	Steady State	0.095 ± 0.197	0.05 ± 0.039	0.039 ± 0.107	0.012 ± 0.045
3	Crisis	0.193 ± 0.152	0.077 ± 0.04	-0.015 ± 0.159	0.046 ± 0.057
	ANOVA p-value	0.0000	0.0084	0.0015	0.0377

Figure 9: Forward 12M Asset Returns by Market Regime - Train

Forward 12M Asset Returns by Market Regime (Test Sample, with ANOVA significance)					
Regime	Regime_Label	Russell 1000	US Agg Bond	Gold	Dollar
0	Mid Expansion	0.121 ± 0.164	-0.011 ± 0.067	0.126 ± 0.151	0.025 ± 0.041
1	Steady State	0.138 ± 0.014	0.05 ± 0.014	0.389 ± 0.026	-0.016 ± 0.019
3	Crisis	0.424 ± 0.092	0.001 ± 0.008	0.033 ± 0.062	-0.068 ± 0.022
	ANOVA p-value	0.0002	0.2597	0.0032	0.0000

Figure 10: Forward 12M Asset Returns by Market Regime - Test

Forward 12M Relative Performance by Market Regime (Train Sample)					
Regime	Regime_Label	Growth – Value (12M fwd)	HY – IG (12M fwd)	Long – Short Treasury (12M fwd)	EM – DM Equities (12M fwd)
0	Mid Expansion	-0.006 ± 0.11	0.028 ± 0.078	0.042 ± 0.082	0.034 ± 0.145
1	Steady State	0.04 ± 0.089	0.011 ± 0.096	0.042 ± 0.075	0.128 ± 0.237
2	Steady State	0.049 ± 0.174	0.007 ± 0.117	0.011 ± 0.074	0.045 ± 0.216
3	Crisis	0.047 ± 0.052	0.019 ± 0.039	0.05 ± 0.064	-0.049 ± 0.123
	ANOVA p-value	0.0013	0.3142	0.0526	0.0000

Figure 11: Forward 12M Relative Performance by Market Regime - Train

Forward 12M Relative Performance by Market Regime (Test Sample)					
Regime	Regime_Label	Growth – Value (12M fwd)	HY – IG (12M fwd)	Long – Short Treasury (12M fwd)	EM – DM Equities (12M fwd)
0	Mid Expansion	0.073 ± 0.151	0.047 ± 0.023	-0.078 ± 0.083	-0.059 ± 0.072
1	Steady State	nan ± nan	nan ± nan	nan ± nan	nan ± nan
3	Crisis	0.063 ± 0.097	0.116 ± 0.033	-0.123 ± 0.032	0.072 ± 0.087
	ANOVA p-value	0.8829	0.0000	0.1899	0.0001

Figure 12: Forward 12M Relative Performance by Market Regime - Test

Key Findings: 1. Crisis Recovery Premium: Crisis regimes consistently deliver the highest forward equity returns (19.3% train, 42.4% test), validating contrarian strategies. The exceptional test premium reflects unprecedented monetary and fiscal intervention.

2. Defensive Asset Restructuring: Bonds transition from strong regime-differentiation ($p = 0.0084$) to insignificance ($p = 0.2597$) as policy-driven rates dominate, while gold reverses from crisis underperformance (-1.5%) to positive safe-haven returns (3.3%), reflecting different crisis dynamics.

3. Factor Rotation Reversal: Growth outperforms during Crisis in training but Value outperforms during Crisis in testing, reflecting the shift from long-duration secular growth (2010s) to value/cyclical leadership during inflation and normalization (2020s).

4. Credit Risk Persistence: High Yield-Investment Grade spreads maintain statistical significance across both periods, confirming credit risk remains a robust regime-dependent factor for portfolio construction.

4.3 Hidden Markov Models

We extract latent macroeconomic regimes through two steps: (1) saliency-based feature reweighting, and (2) estimation of a sticky Gaussian HMM. This structure improves regime separability while remaining stable in high dimensions.

4.3.1 Step 1. Feature-Saliency Reweighting

Following FSHMM, each feature l has a saliency score $\rho_l \in [0, 1]$ that measures how strongly it depends on the regime. In the full FSHMM, the emission density for feature l is

$$p(y_{t,l} | S_t = k) = \rho_l \mathcal{N}(\mu_{k,l}, \sigma_{k,l}^2) + (1 - \rho_l) \mathcal{N}(\mu_{0,l}, \tau_l^2).$$

We estimate saliency parameters on the training window by running EM on this FSHMM.

Instead of dropping low-saliency variables, we construct a weighted feature matrix

$$\tilde{y}_{t,l} = w_l \hat{z}_{t,l}, \quad w_l = \sqrt{\max(\hat{\rho}_l, \varepsilon)},$$

where $\hat{z}_{t,l}$ is the standardized feature l (using training moments), $\hat{\rho}_l$ is the estimated saliency, and ε is a small floor. This retains all variables but downweights those with weak regime sensitivity, stabilizing the HMM fit.

4.3.2 Step 2. Estimation and Sticky Transition Structure

We winsorize each series at 1%–99% and compute z-scores using the training sample only. The resulting matrix Z_{train} is transformed into $\tilde{Z}_{\text{train}}$ using the saliency weights.

For a given K we then estimate a sticky Gaussian HMM on $\tilde{Z}_{\text{train}}$:

- **Emissions.** We use diagonal covariance matrices $\Sigma_k = \text{diag}(\sigma_{k,1}^2, \dots, \sigma_{k,p}^2)$ to reduce parameters and improve robustness.
- **Sticky transitions.** The initial transition matrix is ergodic with high self-transition probabilities, $A_{kk}^{(0)} \approx 0.9$, and a Dirichlet-type prior penalizes very volatile transition patterns. This enforces persistent regimes and rules out unrealistically fast switching.

Parameters are estimated via Baum–Welch EM with multiple initializations. Posterior regime probabilities and the most likely path are then obtained via forward–backward and Viterbi recursions.

4.3.3 Step 3. Selecting the Number of Regimes via BIC

We estimate models with $K \in \{3, 4, 5\}$ on the saliency-weighted training data and choose K using the Bayesian Information Criterion (BIC),

$$\text{BIC}(K) = -2 \log L_K + d_K \log(T),$$

where L_K is the maximized likelihood, d_K the number of free parameters, and T the number of training observations.

In our data the BIC is minimized at $K = 5$, indicating that five regimes provide the best trade-off between fit and parsimony; adding further regimes would mainly add complexity.

4.3.4 Step 4. Regime Decoding

After calibrating the sticky HMM on $\tilde{Z}_{\text{train}}$, we nowcast regimes out-of-sample under a strict no-look-ahead rule:

- **Training path.** The in-sample regime path is the Viterbi path on the full training window.
- **OOS path.** For each OOS month t , we use the entire history up to t (training plus past OOS) and decode the last state in the Viterbi path for $\{1, \dots, t\}$, with all parameters and saliency weights kept fixed at their training estimates.

4.3.5 Results

Feature Saliency. Figure 13 reports the top-30 saliency scores from the training sample.

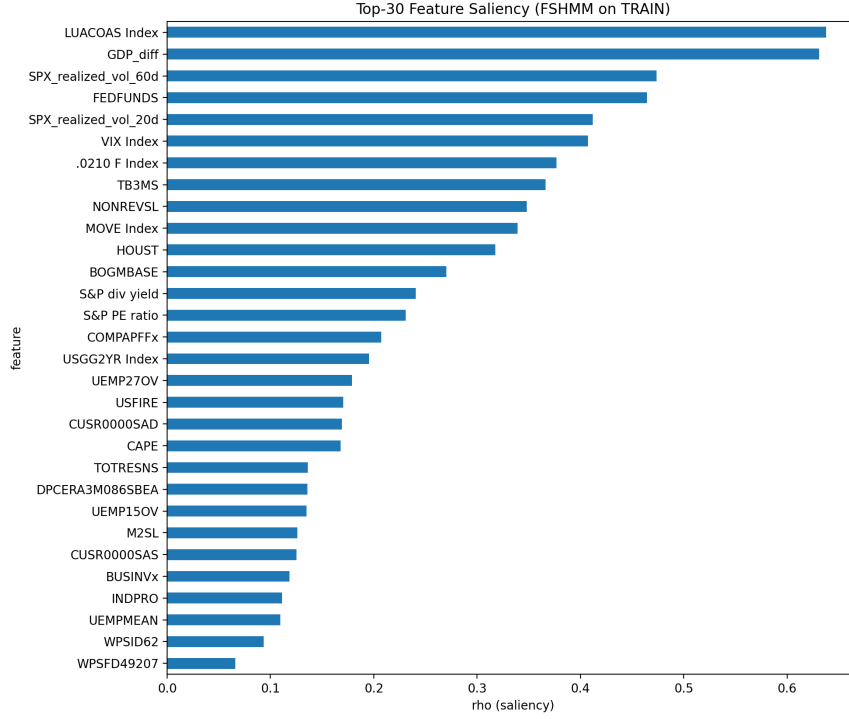


Figure 13: Top-30 feature saliency scores from FSHMM on the training sample.

The highest-ranked variables are broad activity indicators (LUACOAS, GDP growth), monetary policy measures (FEDFUNDS, short rates), and equity–volatility series (realized volatility, VIX)—precisely the indicators that vary most over the business cycle. Structural or demographic variables receive low weights. The saliency pattern confirms that regime differentiation is driven by macro–financially meaningful sources of variation.

Choice of Number of Regimes. Figure 14 shows BIC values for $K = 3, 4, 5$. The criterion decreases monotonically and reaches its minimum at $K = 5$, which we adopt.

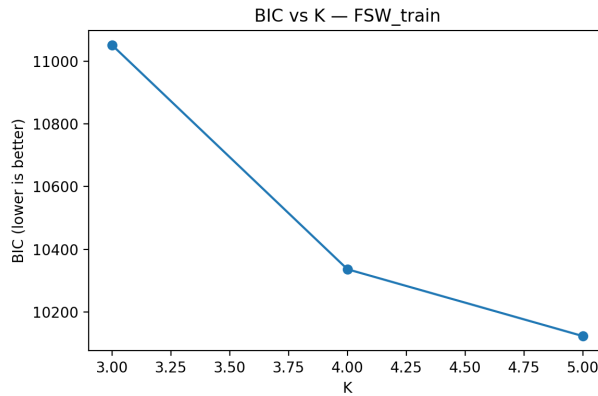


Figure 14: BIC for saliency-weighted HMMs with different numbers of regimes.

Regime Path. Figure 15 presents the full-sample regime sequence. In-sample, the model aligns well with major U.S. macro-financial episodes: early-1990s recession, late-1990s expansion, the dot-com bust, the mid-2000s credit boom, and the 2008 downturn.

Both the 2008 GFC and 2020 COVID shock load into the same high-volatility state (Regime 5), characterized by broad risk aversion and safe-haven strength. Transitions following each crisis move into distinct recovery states, consistent with forward-return patterns.

Overall, the chronology follows macro intuition: expansions and recessions form persistent states, while major dislocations generate sharp regime breaks. Several regimes span multiple years, reflecting both transition persistence and business-cycle dynamics.

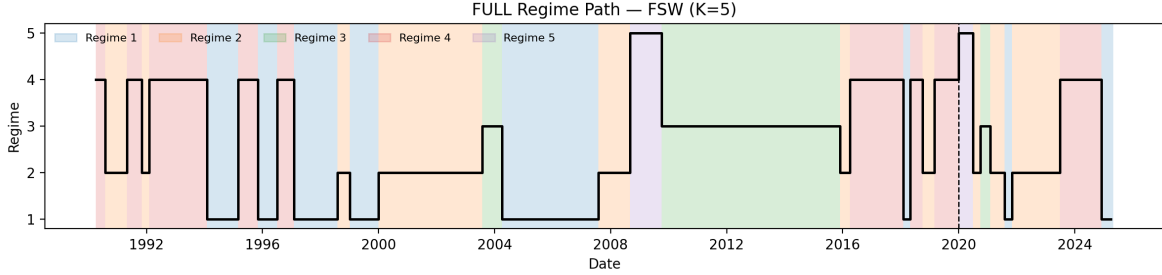


Figure 15: Full-sample HMM regime path ($K = 5$). Dashed line denotes train/OOS split.

Forward-Return Analysis. We evaluate regime separability using 6-month forward absolute returns (equities, bonds, gold, dollar) and relative spreads (Value-Growth, IG-HY, yield-curve slope, DM-EM). A one-way ANOVA tests whether mean forward returns differ across regimes.

Absolute (6M fwd) — Train					
Regime	Russell 1000	US Agg Bond	Gold	Dollar	N
1	0.067 ± 0.063	0.028 ± 0.029	0.034 ± 0.112	0.002 ± 0.038	94
2	-0.009 ± 0.155	0.040 ± 0.028	0.039 ± 0.101	0.013 ± 0.049	82
3	0.053 ± 0.081	0.019 ± 0.021	0.012 ± 0.104	0.015 ± 0.040	82
4	0.051 ± 0.073	0.031 ± 0.034	0.021 ± 0.073	0.024 ± 0.039	86
5	0.115 ± 0.183	0.046 ± 0.017	0.123 ± 0.072	-0.027 ± 0.054	13
ANOVA p-value	0.0000	0.0000	0.0031	0.0001	

Figure 16: Absolute 6-month forward returns by regime (training sample).

Relative (6M fwd) — Train					
Regime	R1000 Value - Growth	IG Corp - HY Corp	Short Tsy - Long Tsy	DM ex-US - EM	N
1	-0.032 ± 0.079	-0.006 ± 0.033	-0.015 ± 0.045	0.006 ± 0.139	94
2	0.026 ± 0.105	0.000 ± 0.094	-0.022 ± 0.051	-0.056 ± 0.118	82
3	-0.008 ± 0.036	-0.010 ± 0.041	-0.034 ± 0.081	0.018 ± 0.060	82
4	-0.023 ± 0.070	-0.005 ± 0.050	-0.020 ± 0.059	-0.039 ± 0.129	86
5	-0.041 ± 0.068	-0.115 ± 0.104	0.012 ± 0.052	-0.148 ± 0.099	13
ANOVA p-value	0.0000	0.0000	0.0686	0.0000	

Figure 17: Relative 6-month forward returns by regime (training sample).

Training-period patterns are strong. Equities outperform in expansionary regimes, bonds and gold in defensive or stress states. Relative spreads behave consistently with macro conditions: Value

outperforms in stronger activity regimes, IG–HY widens in risk-off states, and the curve steepens in recoveries. ANOVA p-values near zero confirm clear statistical separation.

Absolute (6M fwd) — Test					
Regime	Russell 1000	US Agg Bond	Gold	Dollar	N
1	0.009 ± 0.076	-0.017 ± 0.056	0.148 ± 0.080	-0.022 ± 0.050	6
2	0.037 ± 0.107	-0.019 ± 0.043	0.008 ± 0.086	0.010 ± 0.040	29
3	0.192 ± 0.067	-0.013 ± 0.010	-0.017 ± 0.067	-0.013 ± 0.022	4
4	0.099 ± 0.074	0.027 ± 0.021	0.156 ± 0.053	0.017 ± 0.026	17
5	0.186 ± 0.102	0.028 ± 0.017	0.137 ± 0.087	-0.040 ± 0.037	6
ANOVA p-value	0.0005	0.0008	0.0000	0.0081	

Figure 18: Absolute 6-month forward returns by regime (test sample).

Relative (6M fwd) — Test					
Regime	R1000 Value - Growth	IG Corp - HY Corp	Short Tsy - Long Tsy	DM ex-US - EM	N
1	0.014 ± 0.086	-0.023 ± 0.020	0.045 ± 0.047	0.018 ± 0.037	6
2	-0.024 ± 0.118	-0.027 ± 0.026	0.058 ± 0.088	0.038 ± 0.054	29
3	0.065 ± 0.053	-0.056 ± 0.026	0.079 ± 0.051	0.056 ± 0.062	4
4	-0.056 ± 0.040	-0.019 ± 0.010	0.014 ± 0.041	0.011 ± 0.045	17
5	-0.194 ± 0.117	-0.027 ± 0.045	-0.015 ± 0.071	-0.090 ± 0.027	6
ANOVA p-value	0.0005	0.1417	0.0731	0.0000	

Figure 19: Relative 6-month forward returns by regime (test sample).

Out-of-sample, several relationships weaken or reverse. For instance, US Agg Bond earns positive mean returns across all regimes in-sample. But in regime 1, 2, 3, it has negative mean out-of-sample.

Relative spreads highlight the instability: DM–EM remains separated, but IG–HY and the Treasury slope show ANOVA p-values above 0.05, indicating no statistically distinct across regimes.

These reversals and reduced cross-regime differences show that the regime–return mapping does not remain stable in new data, motivating further stability analysis.

Regime Summary & Interpretation. Combining the regime path and return patterns, we interpret the five regimes as follows:

Regime	Market Characteristics (from returns)	Historical Alignment / Interpretation	Label
1	Equities ↑, Bonds ↑, Gold ↑; Dollar flat; Credit moderate; Short Tsy > Long Tsy	Mid-cycle expansions (e.g., 1995–1999, 2004–2007) with stable growth & balanced sentiment	Mid Expansion / Stable Growth
2	Equities flat/↓; Bonds ↑; Gold ↑; HY < IG; Short Tsy > Long Tsy (flattening)	Slowdown periods (early-2000s, 2008 onset, 2022–24 tightening); rising uncertainty, defensive rotation	Slowdown / Risk-Off
3	Equities ↑↑; Bonds flat/↓; Gold ↓; HY > IG; Long Tsy > Short Tsy	Risk-on recoveries (1993–95, post-2010); high liquidity, strong risk appetite	Early Expansion / Risk-On
4	Equities moderate ↑; Bonds mixed; Gold ↑; IG > HY; Short Tsy > Long Tsy	Late-cycle tightening (2016–2018); inflation pressure, policy tightening	Late Expansion / Pre-Stress
5	Equities ↓↓; Gold ↑↑; Dollar ↑↑; IG > HY; Short Tsy >> Long Tsy	Crisis periods (2000 bust, 2008–09, early 2020); liquidity stress, flight to quality	Crisis / Market Stress

Figure 20: HMM Regime Definition Table

Train–Test Stability. Despite strong in-sample structure, regime stability deteriorates out-of-sample. The forward-return analysis already shows reversals and weaker separation, indicating that the learned regime–return mapping does not generalize.

Figure 21 compares the OOS nowcast path (parameters fixed at training estimates) with a full-sample refit, which serves as an oracle. The refit assigns 2008 and 2020 to different regimes, while the frozen model must place both shocks into pre-estimated states, producing classification mismatches and timing errors near regime boundaries.

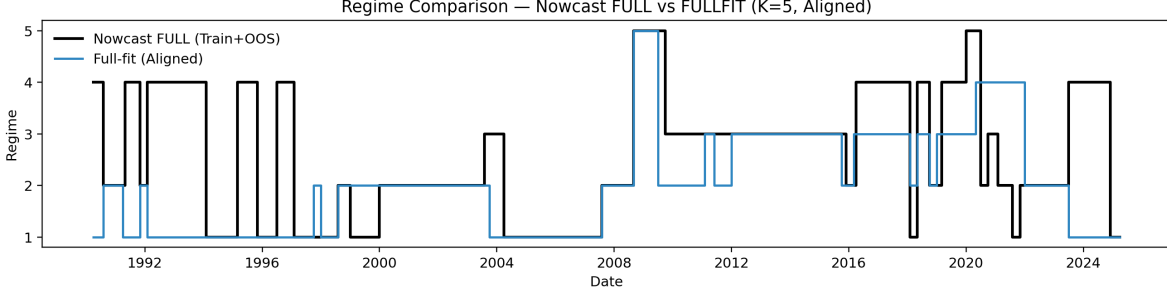


Figure 21: Comparison between OOS nowcast regimes and full-sample refit regimes.

This highlights two limitations of fixed-parameter HMMs: (i) transition and emission relationships estimated in-sample may drift over time, and (ii) the model cannot form new states in response to unprecedented shocks. Consequently, regime identification becomes unstable—especially in periods of structural change.

4.3.6 Discussion

In summary, the HMM is widely used for its clear advantages:

- **Regime evolution:** Unlike clustering methods, it provides a time-ordered regime path and transition structure.
- **Strong regime identification:** It captures economically meaningful states and cleanly differentiates asset-return patterns in-sample.

However, in our fixed-parameter setup, it also faces a key limitation:

- **Weak out-of-sample stability due to short-lived parameter validity:** The transition and emission parameters remain accurate only for a limited horizon, and their effectiveness decays quickly as macro conditions shift, leading to unstable OOS classifications and weakened return patterns.

Overall, the HMM is powerful for retrospective regime analysis, but with fixed parameters its usefulness declines over multi-year forecast windows such as the five-year period studied here.

4.4 Regularized Jump Model

Among several regime-identification methods explored in this study, the Regularized Jump Model (RJM) serves as one of the key candidates due to its ability to incorporate temporal persistence and embedded feature selection within a unified optimization framework. RJM extends the classical Statistical Jump Model by penalizing the number of regime switches while jointly shrinking regime-specific feature means. This yields interpretable, persistent, and high-dimensionality-robust

regime structures. In what follows, we present the model formulation, initialization scheme, calibration algorithm, and hyperparameter tuning procedure based on the elbow method and stationary-bootstrap stability analysis.

4.4.1 Model Definition

Let $y_t \in \mathbb{R}^p$ denote a p -dimensional observation at time t , and let $S = (s_1, \dots, s_T)$ be the latent regime sequence with $s_t \in \{1, \dots, K\}$. The RJM objective takes the form

$$\min_{\Theta, S} \sum_{t=1}^T \|y_t - \theta_{s_t}\|_2^2 + \lambda \sum_{t=1}^{T-1} \mathbf{1}\{s_t \neq s_{t+1}\} + \gamma \sum_{j=1}^p \|\mu_{\cdot, j}\|_2, \quad (28)$$

where θ_k is the centroid of regime k , and $\mu_{\cdot, j} = (\mu_{1, j}, \dots, \mu_{K, j})^\top$ denotes the vector of regime-specific means for feature j . The group-lasso penalty $\|\mu_{\cdot, j}\|_2$ induces feature selection jointly across regimes, stabilizing inference in high-dimensional settings.

4.4.2 Initialization

We follow a two-stage deterministic initialization procedure:

1. **Initial regime labels.** Perform K -means clustering on $\{y_t\}_{t=1}^T$ to obtain an initial sequence $S^{(0)}$, which avoids sensitivity to random seeds and provides a geometry-based starting point.
2. **Initial centroids.** For each regime k , compute

$$\theta_k^{(0)} = \frac{1}{|C_k|} \sum_{t: s_t^{(0)} = k} y_t,$$

where C_k is the set of indices assigned to regime k .

4.4.3 Calibration via Coordinate Descent

RJM is optimized by alternating between updating the regime parameters Θ and the state sequence S [3]. The complete procedure is summarized in Algorithm 2.

Algorithm 2 Calibration of the Regularized Jump Model

- 1: **Input:** Data $\{y_t\}$, number of regimes K , penalties (λ, γ) .
- 2: **Initialize:** $(S^{(0)}, \Theta^{(0)})$ via K -means.
- 3: **for** $m = 0, 1, 2, \dots$ **until** convergence **do**
- 4: **(Parameter update)** For each regime k , compute the unregularized centroid:

$$\tilde{\theta}_k = \frac{1}{|C_k|} \sum_{t: s_t^{(m)} = k} y_t.$$

- 5: Apply group-lasso shrinkage to each feature dimension:

$$\mu_{\cdot, j}^{(m+1)} = \left(1 - \frac{\gamma}{\|\tilde{\mu}_{\cdot, j}\|_2}\right)_+ \tilde{\mu}_{\cdot, j},$$

yielding $\Theta^{(m+1)}$.

- 6: **(State update)** Solve

$$S^{(m+1)} = \arg \min_S \sum_t \|y_t - \theta_{s_t}^{(m+1)}\|_2^2 + \lambda \sum_{t=1}^{T-1} \mathbf{1}\{s_t \neq s_{t+1}\},$$

via dynamic programming:

$$C_t(k) = \|y_t - \theta_k^{(m+1)}\|_2^2 + \min \left(C_{t-1}(k), \min_{k' \neq k} C_{t-1}(k') + \lambda \right).$$

- 7: **if** $S^{(m+1)} = S^{(m)}$ **then**
 - 8: **break**
 - 9: **end if**
 - 10: **end for**
 - 11: **Output:** Estimated regimes $\hat{S}$ and centroids $\hat{\Theta}$. =0
-

4.4.4 Selecting the Number of Regimes via the Elbow Method

Because the Jump Model penalizes regime switches, the objective $\mathcal{L}(K)$ is not guaranteed to decrease monotonically as K increases. Adding regimes lowers within-state error but may introduce extra transitions, raising the jump-penalty term. Consequently, $\mathcal{L}(K)$ may flatten or even rise with larger K .

To choose a suitable number of regimes, we therefore apply an elbow-type diagnostic to the sequence $\mathcal{L}(K)$, focusing on the marginal improvement $\Delta\mathcal{L}(K)$. The preferred value of K is where reductions in the penalized loss diminish noticeably and the resulting regime sequence remains economically interpretable.

In our empirical application, the penalized-loss curve levels off around $K = 4$, indicating that this specification balances improved fit with avoiding excessive temporal fragmentation.

4.4.5 Hyperparameter Tuning via Stationary Bootstrap and Stability

After fixing a candidate K , we tune (λ, γ) using a stability metric evaluated under stationary bootstrap resampling.

Stationary Bootstrap Because macro-financial time series exhibit autocorrelation, standard bootstrap methods are inappropriate. We therefore adopt the stationary bootstrap of Politis and Romano (1994) [6], which preserves the data’s dependence structure. Bootstrap samples are constructed by concatenating blocks whose lengths follow a geometric distribution with mean $1/p$:

- With probability p , start a new block from a randomly sampled index.
- Otherwise, continue the current block by copying the next observation.

This procedure produces bootstrap sequences $\{y_t^*\}$ suitable for evaluating model robustness under dependent data.

Stability Criterion For each candidate (K, λ, γ) :

1. Generate B stationary-bootstrap samples.
2. Fit RJM to each sample, yielding regime sequences $S^{*(b)}$.
3. Compute stability relative to the original estimate:

$$\text{Stability} = \frac{1}{B} \sum_{b=1}^B \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{S_t^{*(b)} = S_t^{\text{orig}}\}.$$

4. Select the (λ, γ) pair that maximizes stability while retaining meaningful sparsity and avoiding degenerate one-regime solutions.

We use the stability defined above as the target function and look for best parameter set with the help of Optuna package[5]. This stability-guided selection ensures that the final regime structure is robust, economically interpretable, and not overly sensitive to sampling variation.

4.4.6 Results

Identified Regime Structure To assess the stability of the Regularized Jump Model (RJM) regime classification, we compare the regime timeline estimated on the train–test split (Figure 22) with the regime timeline estimated on the full sample (Figure 23). The two results exhibit highly consistent structure: regime blocks occur at nearly identical locations, and the model preserves the macroeconomic interpretation of each regime even when additional data are included.

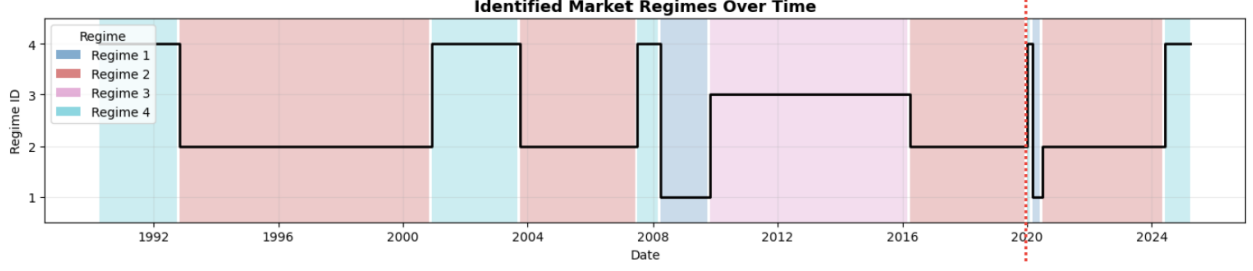


Figure 22: RJM regimes estimated using the train–test split (1989–2019 training, 2020–2024 test). The vertical dotted line marks the train–test boundary.

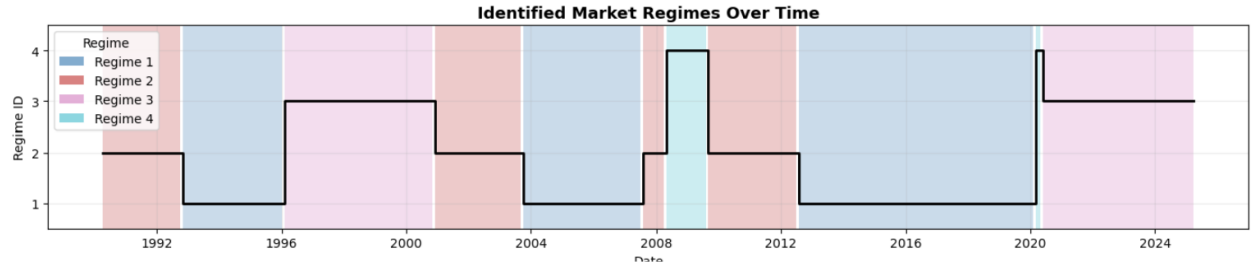


Figure 23: RJM regimes estimated using the full 1990–2024 sample. The regime structure remains highly stable when re-estimated with additional data.

Comparing Figures 22 and 23, the overall regime segmentation is remarkably stable. Long expansions, contraction phases, and recovery periods appear in nearly the same locations across both estimation settings. The inclusion of post-2020 data slightly shifts the boundaries of the most recent regimes but does not alter their economic meaning. This stability is crucial for real-time applications: a model that produces consistent regime identities across different information sets is far more reliable for forward-looking asset allocation.

To quantify this correspondence formally, Table 13 reports the mapping between regimes in the train–test estimation and the full-sample estimation. For example, train–test Regime 1 aligns most closely with full-sample Regime 4, as both exhibit prolonged late-cycle characteristics with tightening credit conditions and slowing industrial activity.

Table 13: Mapping Between RJM Regimes: Train–Test Estimation vs. Full-Sample Estimation

Train–Test Regime	Matching Full-Sample Regime
1	4
2	1, 3
3	1, 2
4	2

Overall, the alignment of regime identities across Figures 22 and 23, reinforced by the mapping in Table 13, demonstrates that the Regularized Jump Model captures persistent and interpretable

macroeconomic structure rather than sample-specific noise. This strong temporal robustness motivates its use as a foundation for systematic multi-asset allocation.

Economic Interpretation of Regimes Figure 24 shows the top 20 selected features, and Table 14 summarizes the macroeconomic interpretation of each regime based on feature profiles, historical timing, and typical market conditions.

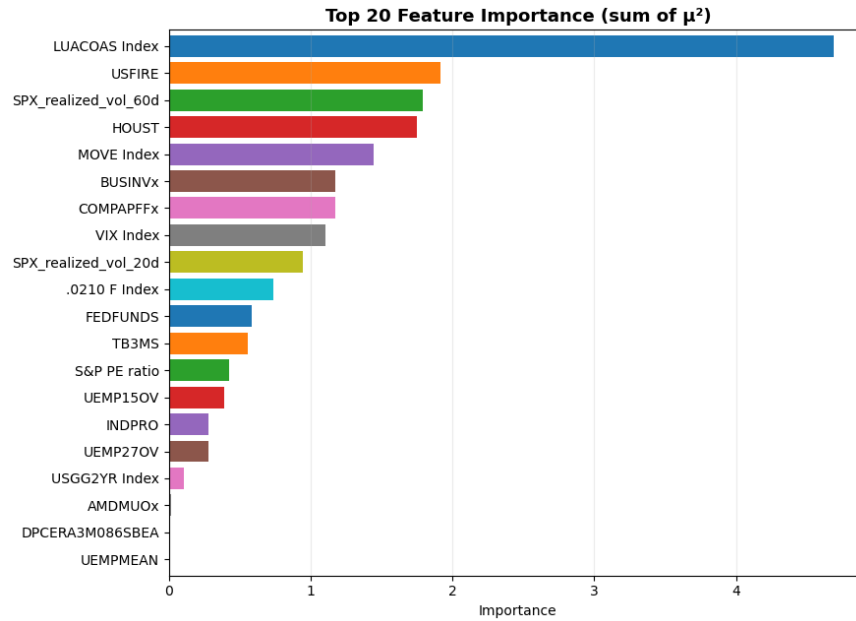


Figure 24: Top 20 important features selected by the RJM.

Regime	Economic Characterization	Dominant Features	Asset Behavior
1	Stable expansion, low stress	LUACOAS, USFIRE, HOUST; low realized vol.; low VIX	Equities strong; credit performs well
2	Late-cycle tightening; rising uncertainty	Interest-rate variables (FEDFUNDS, TB3MS); softening activity; elevated VIX/MOVE	Gold strengthens; equities weaken
3	Crisis regime; severe macro stress	High realized vol. (20–60d); elevated VIX/MOVE; falling real-activity indicators	Gold outperforms; bonds strong; USD strengthens
4	Post-crisis rebound; unstable recovery	Moderating volatility; mixed real-activity signals	Equities recover with high volatility

Table 14: Economic interpretation of RJM regimes (train-test split).

The feature selection pattern confirms that the RJM emphasizes macroeconomic activity, monetary-policy conditions, and volatility—variables commonly associated with market phases.

Regime Separability via Forward Returns We evaluate regime separability using forward 12-month returns across major asset classes. Tables 15-16 summarize the sample means, standard

deviations, and annualized volatilities across regimes, alongside ANOVA p -values.

Cross-Asset Separability To evaluate whether the inferred regimes correspond to economically distinct environments, we examine forward 12-month returns across major asset classes, including U.S. equities, investment-grade bonds, gold, and the U.S. dollar. As shown in Table 15, broad asset classes exhibit clear and statistically significant differences across regimes. Equity markets display strong performance in expansionary regimes and weaker or negative outcomes in stress regimes, while safe-haven assets such as investment-grade bonds and gold outperform during crisis periods. These results demonstrate that the Regularized Jump Model effectively captures meaningful macro-financial state variation and provides a strong basis for top-down asset allocation.

Table 15: Forward 12-month returns across major asset classes by regime. RJM shows strong cross-asset separability.

Asset Class	Regime 1	Regime 2	Regime 3	Regime 4
Russell 1000	0.110 ± 0.093	0.043 ± 0.179	0.146 ± 0.174	0.093 ± 0.319
ANOVA p		0.0000		
US Agg Bond	0.043 ± 0.043	0.080 ± 0.039	0.034 ± 0.068	0.071 ± 0.034
ANOVA p		0.0000		
Gold	0.074 ± 0.159	0.098 ± 0.140	0.025 ± 0.157	0.173 ± 0.138
ANOVA p		0.0001		
Dollar	0.021 ± 0.059	0.011 ± 0.060	0.035 ± 0.051	0.001 ± 0.083
ANOVA p		0.0089		

Within-Asset-Class Separability In contrast to the strong cross-asset patterns, the regimes display only limited explanatory power for within-asset-class differences. Table 16 summarizes forward 12-month spreads across equity styles (Value vs. Growth) and regional equity benchmarks (U.S., DM ex-US, EM). Across all cases, regime-driven differences are small and statistically insignificant.

This asymmetry indicates that RJM captures macroeconomic risk conditions that matter for broad asset allocation but does not generate strong cross-sectional signals within asset classes. As such, the inferred regimes should be used primarily for top-down asset allocation rather than style or regional equity selection.

Table 16: Forward 12-month relative return spreads by regime (selected spreads).

Spread	Regime 1	Regime 2	Regime 3	Regime 4
Equity Style & Region				
Growth – Value	0.001 ± 0.017	0.000 ± 0.030	0.005 ± 0.041	0.014 ± 0.031
ANOVA p		0.1302		
EM – DM Equities	0.001 ± 0.039	0.005 ± 0.050	-0.009 ± 0.053	0.011 ± 0.060
ANOVA p		0.0863		
Credit Spreads				
HY – IG	0.003 ± 0.021	0.007 ± 0.035	-0.006 ± 0.037	0.025 ± 0.058
ANOVA p		0.0001		
Bond Curve				
Long – Short Treasury	-0.004 ± 0.042	0.008 ± 0.061	-0.012 ± 0.046	0.036 ± 0.109
ANOVA p		0.0006		

4.4.7 Discussion

The Regularized Jump Model exhibits several desirable properties:

- **Robust regime identification:** long and economically coherent segments.
- **Clear feature selection:** sparse, interpretable macro drivers.
- **Strong cross-asset separability:** distinct return behavior across major asset classes.

However, RJM shows limited ability to differentiate:

- style-level equity spreads,
- regional equity differences,
- finer bond-segment relative performance.

Thus, RJM is highly effective for macro regime identification and top-down allocation, but less suited for within-asset-class cross-sectional strategy design.

5 Regime-Aware Tactical Asset Allocation Framework

After identifying macroeconomic regimes, we construct a tactical asset allocation framework that adjusts portfolio weights based on the prevailing regime signals. The objective is to link each regime to relative asset class performance and allocate capital toward assets that historically perform better under similar macroeconomic conditions.

The framework evaluates regime-specific return profiles and risk measures across major asset classes and uses these regime labels to inform tactical tilts. This allows the portfolio to adapt dynamically to changing macroeconomic environments while maintaining a systematic and quantitatively justified allocation process.

5.1 Benchmark

The benchmark portfolio serves as the strategic reference point for all tactical allocation models evaluated in this project. It follows a **60/40 equity-bond allocation** provided by Vanguard, combining U.S. equities with high-quality fixed-income exposure. This structure offers diversified exposure across growth, duration, and credit premia and represents the neutral investment stance against which all regime-aware strategies are evaluated.

Table 17: Asset Universe and Benchmark Weights

Asset Class	Weight	Category
Russell 1000 Value	25%	Large Cap Equity
Russell 1000 Growth	25%	Large Cap Equity
Russell 2000	10%	Small Cap Equity
US Short-term Treasury	10%	Government Bonds
US Long-term Treasury	10%	Government Bonds
US IG Corporate Bond	15%	Credit
US HY Corporate Bond	5%	Credit

This benchmark portfolio reflects a balanced and institutionally relevant mix of risk factors and asset classes. All subsequent performance, attribution, and risk-adjusted comparisons in this study are evaluated relative to this 60/40 strategic allocation.

5.2 Relative Performance Based Asset Allocation

In addition to regime identification, we implement a systematic signal-driven allocation framework that adjusts portfolio weights based on the relative performance of key market premia. The framework is designed to tilt away from the strategic benchmark in a disciplined manner, using only economically interpretable signals and a tightly controlled tilt budget. Consistent with the rest of the paper, all predictive signals use a **6-month forward-looking return**.

Step 1: Measuring Relative Performance Signals

For each month, we compute the relative performance of five market premia:

1. **Equity Risk Premium:** equities vs. short-term Treasuries
2. **Value Premium:** value vs. growth equities
3. **Credit Premium:** high-yield vs. investment-grade credit
4. **Term Premium:** long-term vs. short-term Treasuries
5. **Size Premium:** small-cap vs. large-cap equities

Positive values indicate relative outperformance of the first asset in each pair; negative values indicate the opposite. These raw factor signals serve as the inputs to the tilt-engine.

Step 2: Allocating the Tilt Budget Using Softmax Weighting

To express tactical views in the portfolio, we allocate a fixed **20% tilt budget** that governs how far weights may deviate from the benchmark. We transform the factor signals using a softmax normalization:

$$\omega_k = \frac{\exp(s_k)}{\sum_{j=1}^5 \exp(s_j)}, \quad \sum_k \omega_k = 1.$$

This produces a smooth ranking where stronger signals receive larger allocations of the tilt budget, while still allowing negative signals to adjust direction.

Each premium receives a tilt magnitude:

$$b_k = 0.20 \times \omega_k.$$

These tilts are then mapped into asset-level adjustments:

- A positive equity risk premium increases allocations to R1000 Value, R1000 Growth, and Russell 2000.
- A positive value–growth premium tilts toward value and away from growth.
- A positive size premium increases small-cap exposure.

- A positive credit premium increases high-yield at the expense of IG credit.
- A positive term premium increases long-duration bonds.

The result is a **signal-based target weight vector** prior to portfolio feasibility constraints.

Step 3: Constrained Optimization to Obtain Implementable Weights

Raw tilt targets may violate investment constraints; therefore we solve a constrained optimization problem that finds the feasible portfolio closest to the desired tilts.

Objective. We minimize deviation from both the signal-based target and the original benchmark:

$$\min_w \|w - w_{\text{target}}\|^2 + \lambda \|w - w_{\text{base}}\|^2 + \gamma \sum_i |w_i - w_{\text{base},i}|.$$

Hard Constraints.

- **No short exposure:** $w_i \geq 0$
- **Full investment:** $\sum_i w_i = 1$
- **Equity bounds:** $0.50 \leq w_{\text{equity}} \leq 0.70$
- **Max deviation from benchmark:** Each asset may move at most ± 10 percentage points from its base weight

The optimization ensures realistic and implementable tactical allocations while preserving consistency with benchmark exposures and risk controls.

Step 4: Performance of Tactical Allocation Across Regime Models

We embed this allocation framework inside three regime-classification models: Hidden Markov Model (HMM), K-means, and Gaussian Mixture Model (GMM). Each model generates monthly regime labels, which govern when the portfolio is rebalanced into the tactical weights. All models use the same 6-month forward return signals in the tilt engine.

Table 18 summarizes the performance of the tactical strategy under each regime model. More details on the performance charts and weight distributions of each model can be found in the Appendix 8.

Model	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	Excess Ret.	Track. Err.	IR
HMM (6M)	6.93%	13.85%	0.501	22.24%	-0.12%	1.06%	-0.11
GMM (6M)	7.34%	14.23%	0.516	23.84%	0.29%	0.89%	0.33
K-Means (6M)	7.83%	13.93%	0.562	22.87%	0.78%	0.94%	0.83
Static Benchmark	7.05%	13.48%	0.523	22.28%	—	—	—

Table 18: Performance comparison of tactical allocations under three regime-classification models.

Notes: Excess Return is computed relative to the static benchmark. Tracking Error is the standard deviation of active returns. The Information Ratio (IR) is defined as Excess Return divided by Tracking Error.

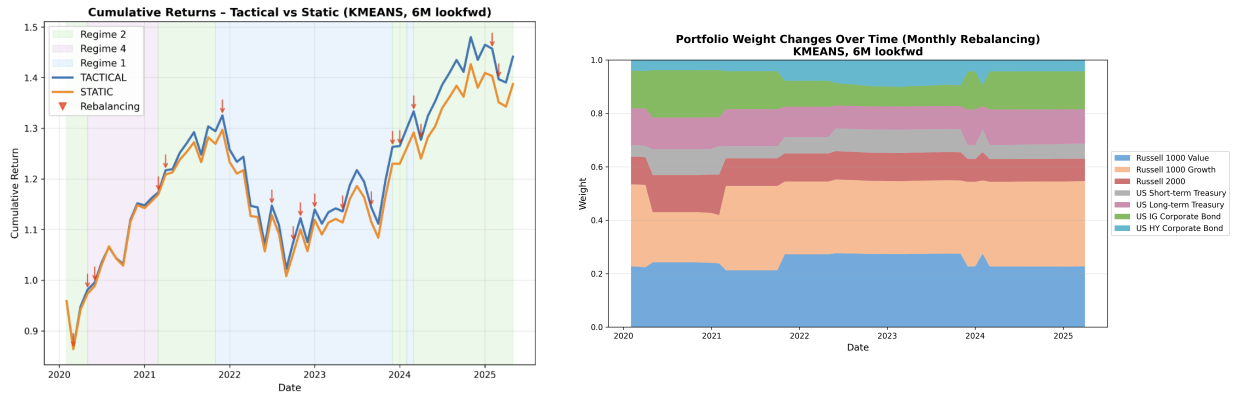


Figure 25: TS K-Means Strategy Over Test Period

Conclusion. The results in Table 18 indicate that the **Time-Series K-Means** model provides the most effective regime classification for driving tactical tilts in this framework. It achieves the **highest annual return (7.83%)**, the **highest Sharpe ratio (0.562)**, and the **strongest Information Ratio (0.83)** among all models, while maintaining volatility and drawdown levels comparable to the benchmark. This outperformance stems from K-Means’ ability to **separate market states based purely on empirical patterns in the data**, enabling the portfolio to allocate risk **at the right time**: increasing equity and credit exposure during improving market conditions and scaling back during weaker environments.

In contrast, the **HMM** model exhibits slower regime transitions and greater sensitivity to noise, leading to negative excess returns and a negative IR. The **GMM** model offers moderate improvements but still falls short of the allocation timeliness achieved by K-Means. Overall, these findings suggest that a **non-parametric, distance-based clustering approach** is better suited for identifying actionable market regimes in this setting, resulting in more effective tactical adjustments and superior risk-adjusted performance.

5.3 Regime-Based Mean–Variance Optimization (MVO)

Once regime labels are identified, we build a tactical allocation framework that maps each regime to a distinct set of portfolio weights. The goal is to tilt the portfolio toward asset classes that historically outperform in the prevailing macro environment, while preserving disciplined risk control through mean–variance optimization.

5.3.1 Overview of the Framework

The procedure follows a train–test structure:

- **Training phase (up to 2019-12):** Using the full training sample, we estimate regime-conditional mean returns and covariance matrices for each regime. A separate MVO problem is solved for every regime, producing one fixed weight vector per regime.
- **Test phase (from 2020-01):** At each month, the nowcasted regime determines which set of return–risk estimates to use. The estimates are updated with an expanding-window (train + past test data) and the portfolio is rebalanced monthly using the same constrained MVO formulation.

This structure allows the portfolio to respond dynamically to changing regimes while ensuring that all decisions rely only on information available at the time.

5.3.2 Training Phase: Regime-Conditional Estimates

For each regime r , we compute:

$$\mu_r = \text{mean returns under regime } r, \quad \Sigma_r = \text{covariance matrix under regime } r.$$

We then solve a constrained mean–variance optimization:

$$\begin{aligned} & \max_w \quad w^\top \mu_r - \gamma w^\top \Sigma_r w \\ \text{s.t.} \quad & \sum_i w_i = 1 \quad (\text{budget constraint}) \\ & w_i \geq 0 \quad (\text{no shorting}) \\ & |w_i - w_i^{\text{bm}}| \leq 0.05 \quad (\text{single-asset tracking band}) \\ & \text{asset-class deviations} \leq 10\% \text{ relative to benchmark weights} \end{aligned}$$

The resulting w_r is the regime-specific tactical portfolio for regime r . These weights are applied to all training months to ensure consistency with the in-sample evaluation.

5.3.3 Test Phase: Expanding-Window, Regime-Conditioned MVO

From 2020-01 onward, portfolio weights are updated in real time:

- At month t , the regime model provides a regime label $\hat{r}_t$.
- Using all data available up to t , we recompute $\mu_{\hat{r}_t}(t)$ and $\Sigma_{\hat{r}_t}(t)$ with an expanding window.

- We solve the same constrained MVO problem as above to obtain the new weight vector w_t .

Rebalancing occurs monthly, allowing the portfolio to adjust its stance whenever the regime shifts or the estimated return–risk profile changes.

5.3.4 MVO Benchmarks

To isolate the value added by regime conditioning, we compare the strategy against:

- **Plain MVO:** identical constraints and estimation window, but using unconditional (regime-agnostic) means and covariances.
- **60/40 Strategy:** the common benchmark as mentioned in Section 5.1

These benchmarks allow us to evaluate whether regime-aware return–risk estimation improves tactical allocation performance relative to standard approaches.

5.3.5 Empirical Performance Across Regime Models

We applied regime-based MVO on all of the HMM, KMeans, and GMM regime path, and summarized their out-of-sample performance as the following table:

Method	Annual Ret	Sharpe	Max Drawdown	Tracking Error	IR	Excess Return
HMM	8.4242%	0.6789	-21.8262%	2.0998%	0.2157	0.4529%
K-Means	8.1603%	0.5951	-22.5446%	1.4013%	0.1349	0.1890%
GMM	7.1751%	0.5227	-24.2099%	1.2557%	-0.6340	-0.7962%
60/40	7.9712%	0.6110	-22.2792%	-	-	-

Table 19: Test-period performance of Regime-MVO portfolios across regime models.

HMM exhibits the strongest overall test-period performance, with superior risk-adjusted returns, drawdown behavior, and excess-return generation relative to K-Means and GMM. We therefore focus on the HMM-based Regime MVO results in the following analysis; full performance tables for the alternative specifications are provided in the Appendix.

HMM-Based Allocation Behaviour. Figure 26 shows the regime-based weights during the test period.

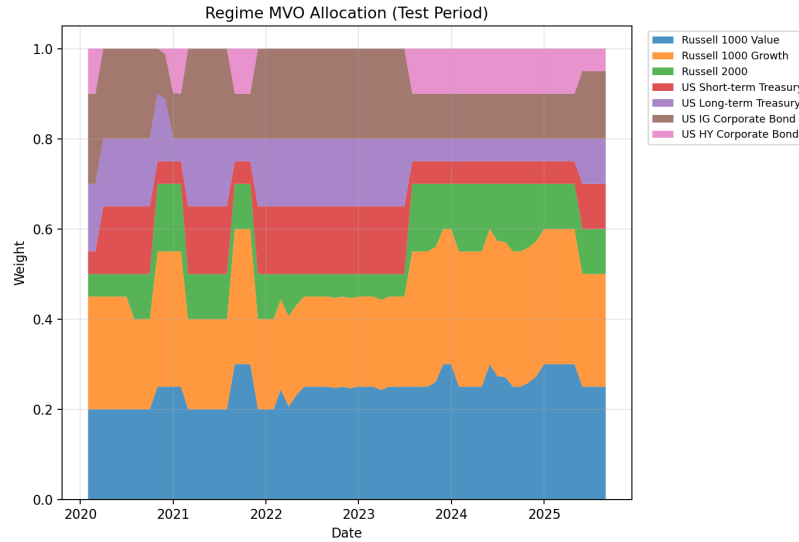


Figure 26: HMM: Regime-based MVO allocations during the test period.

We can see that the portfolio leans into equities in expansionary regimes and shifts toward Treasuries and IG credit in stress regimes (Regime 5). These tilts move smoothly with the regime labels and provide a concrete link between macro signals and asset-class exposure.

Cumulative Performance. Figure below compares cumulative returns for Regime MVO, Plain MVO, and the 60/40 benchmark. The left panel shows the full sample; the right panel focuses on the test period.

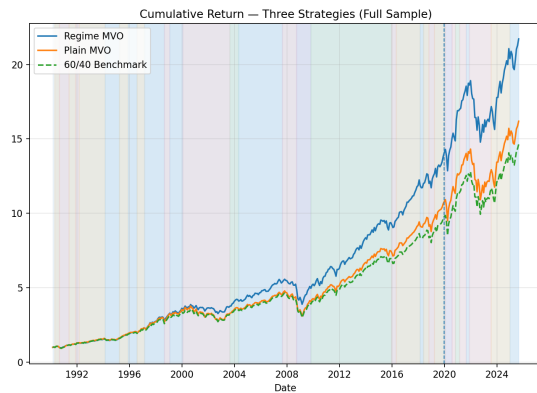


Figure 27: Full sample

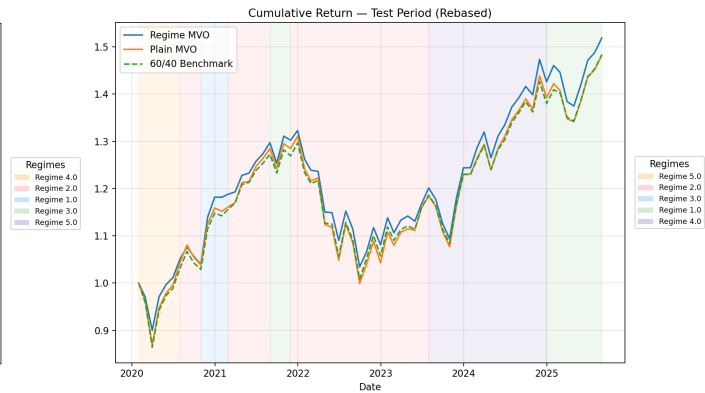


Figure 28: Test period (rebased)

In the full sample (left panel), Regime MVO clearly dominates: it avoids part of the drawdowns around the GFC and COVID and compounds faster in subsequent recoveries. In the test period (right panel), all three strategies move closely together, but Regime MVO keeps a small, persistent lead over Plain MVO and the 60/40 benchmark.

Test-Period Diagnostics. Figures below summarize test-period behaviour using rolling excess return, rolling information ratio, and drawdowns.

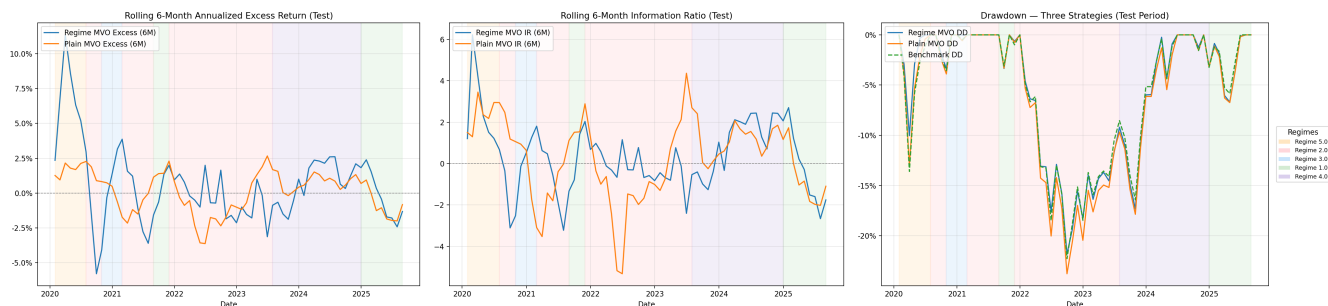


Figure 29: Excess Return

Figure 30: Information Ratio

Figure 31: Maximum Drawdown

The left panel shows that Regime MVO delivers higher 6-month annualized excess returns in stress periods such as early 2020 and mid-2022, when the regime model tilts toward defensive assets. The middle panel indicates that its rolling information ratio is more stable: Plain MVO experiences deep negative IR episodes, while Regime MVO stays closer to zero or positive territory. The right panel shows that drawdowns for Regime MVO are slightly shallower, especially around crisis regimes, confirming that regime conditioning provides some downside protection.

Performance Summary. The following table provide a quantified result of HMM-based MVO portfolio performance.

	Full (Regime MVO)	Full (Plain MVO)	Full (Benchmark)	Train (Regime MVO)	Train (Plain MVO)	Train (Benchmark)	Test (Regime MVO)	Test (Plain MVO)	Test (Benchmark)
Annual Return	0.091478815	0.083827517	0.080625814	0.092849592	0.084397861	0.080798833	0.084241922	0.080816439	0.079712379
Annual Volatility	0.096486308	0.102813198	0.099446133	0.090307423	0.09555284	0.092405135	0.124087843	0.134807672	0.130464586
Sharpe (rf=0)	0.948101516	0.815338099	0.810748615	1.028150168	0.883258528	0.874397647	0.6788894	0.599494361	0.610988632
Max Drawdown	-0.301093028	-0.353567952	-0.352765824	-0.301093028	-0.353567952	-0.352765824	-0.218261721	-0.237834415	-0.222792269
Tracking Error	0.016889903	0.007023804	-	0.015969886	0.006438565	-	0.020997867	0.00951276	-
Information Ratio	0.642573298	0.45583604	-	0.754592674	0.558979849	-	0.215714409	0.116060952	-
Excess Return	0.010853001	0.003201703	-	0.012050759	0.003599028	-	0.004529542	0.00110406	-

Figure 32: Performance summary for HMM-based AA

The regime-based MVO consistently outperforms both plain MVO and the 60/40 benchmark. It earns higher annual returns, maintains a better Sharpe ratio, and experiences a smaller test-period drawdown. Information ratios are stronger in both full and test samples, while excess returns remain positive. Overall, regime conditioning improves risk-adjusted performance even out-of-sample.

5.4 Regime-Aware Benchmark-Tilt (RABT) Strategy

The Regime-Aware Benchmark-Tilt (RABT) strategy introduces systematic, regime-conditioned deviations from the benchmark allocation while enforcing tight implementation constraints to ensure interpretability and real-time feasibility. Rather than allocating directly based on regime labels, the strategy uses regime-conditional forward return profiles to generate relative signals that tilt the portfolio around the benchmark in a disciplined manner. All decisions are based solely on information available at the time, and the framework fully avoids lookahead bias.

5.4.1 Tradable Regime Signal

After regimes are identified from the time-series clustering model, each month-end regime label is shifted forward by one calendar month to form a tradable signal. Let r_t denote the regime assigned to the window ending at time t . The strategy uses the lagged regime

$$s_{t+1} = r_t,$$

ensuring that the regime used for allocation at the start of month $t + 1$ is based exclusively on past information. This one-month lag eliminates implicit lookahead that would otherwise arise from clustering performed on full-window observations.

5.4.2 Forward-Return Profiling

For each asset i and date t , the six-month forward return is computed as

$$F_{i,t}^{(6M)} = \frac{P_{i,t+126}}{P_{i,t}} - 1,$$

where 126 trading days approximates six months. Conditional on regime $s_t = k$, we estimate regime-specific forward-return means

$$\mu_{i,k} = \mathbb{E}\left[F_{i,t}^{(6M)} \mid s_t = k\right].$$

Because regimes may differ in sample size, raw regime means can be unstable. We therefore apply a shrinkage adjustment toward the global mean $\bar{\mu}$:

$$\tilde{\mu}_{i,k} = w_k \mu_{i,k} + (1 - w_k) \bar{\mu}, \quad w_k = \frac{n_k}{n_k + \lambda \bar{n}},$$

where n_k is the number of observations in regime k , and λ controls shrinkage intensity. Well-sampled regimes remain regime-specific, whereas sparse regimes are pulled toward the global average to reduce noise.

5.4.3 Portfolio Constraints

Portfolio weights are constructed relative to the benchmark w^{bench} . Let w_t denote the portfolio weight vector at time t . The strategy enforces:

$$\begin{aligned} \text{(i) No shorting:} & \quad w_{i,t} \geq 0, \\ \text{(ii) Full investment:} & \quad \sum_i w_{i,t} = 1, \\ \text{(iii) Asset-level tilt limits:} & \quad |w_{i,t} - w_i^{\text{bench}}| \leq 0.05, \\ \text{(iv) Group-level tilt limits:} & \quad \left| \sum_{i \in G} w_{i,t} - \sum_{i \in G} w_i^{\text{bench}} \right| \leq 0.10, \end{aligned}$$

for each asset class group $G \in \{\text{Equity, Rates, Credit}\}$.

These constraints ensure diversification, realistic turnover, and tight tracking relative to the benchmark.

5.4.4 Signal-to-Weight Mapping

For the active regime $s_t = k$, we compute standardized signals within each asset class group:

$$z_{i,k} = \frac{\tilde{\mu}_{i,k} - \tilde{\mu}_{G,k}}{\sigma_{G,k}}, \quad i \in G,$$

where $\tilde{\mu}_{G,k}$ and $\sigma_{G,k}$ are the mean and standard deviation of the shrunk returns within group G . These standardized signals are translated into target tilts relative to the benchmark via a constrained quadratic program. Let w^{des} denote the target-tilt vector:

$$w^{\text{des}} = w^{\text{bench}} + \alpha \frac{z}{\sum_i |z_i|}, \quad \alpha = 0.05.$$

The implemented weights solve:

$$\min_w \|w - w^{\text{des}}\|_2^2 \quad \text{subject to the constraints above.}$$

This formulation ensures smooth allocations that remain responsive to signals while respecting risk and diversification limits.

5.4.5 Rebalancing

The portfolio rebalances monthly on the first business day. For each rebalance date t , the procedure is:

1. Read the lagged regime signal s_t .
2. Compute regime-conditional shrunk forward returns $\tilde{\mu}_{\cdot, s_t}$.
3. Construct standardized signals z_{i, s_t} within each asset class group.
4. Solve the quadratic program to obtain updated tilts.

This schedule provides a realistic implementation frequency and allows the portfolio to adjust promptly to regime transitions without incurring excessive turnover.

5.4.6 Algorithm

Algorithm 3 summarizes the full RABT procedure.

Algorithm 3 Regime-Aware Benchmark-Tilt Strategy

- 1: **Input:** Benchmark weights w^{bench} ; lagged regime signal s_t ; six-month forward returns $\{F_{i,t}^{(6M)}\}$.
- 2: **Regime-conditional means:** For each regime k , compute

$$\mu_{i,k} = \mathbb{E}\left[F_{i,t}^{(6M)} \mid s_t = k\right], \quad n_k = \#\{t : s_t = k\}.$$

- 3: **Shrinkage adjustment:** For each asset i and regime k ,

$$\tilde{\mu}_{i,k} = \lambda_k \mu_{i,k} + (1 - \lambda_k) \bar{\mu}, \quad \lambda_k = \frac{n_k}{n_k + \rho},$$

where $\bar{\mu}$ is the global mean and ρ is a ridge parameter.

- 4: **Standardized signals within groups:** For active regime s_t , compute

$$z_{i,s_t} = \frac{\tilde{\mu}_{i,s_t} - \bar{\mu}_G}{\sigma_G + 10^{-8}}, \quad i \in G.$$

- 5: **Desired tilted weights:**

$$w^{\text{des}} = w^{\text{bench}} + 0.05 \frac{z}{\sum_i |z_i| + 10^{-8}}.$$

- 6: **Optimization step:** Solve

$$\begin{aligned} \min_w \quad & \|w - w^{\text{des}}\|_2^2 \\ \text{s.t.} \quad & w_i \geq 0, \quad \sum_i w_i = 1, \\ & |w_i - w_i^{\text{bench}}| \leq 0.05, \quad \forall i, \\ & \left| \sum_{i \in G} w_i - \sum_{i \in G} w_i^{\text{bench}} \right| \leq 0.10, \quad \forall G. \end{aligned}$$

- 7: **Output:** Updated portfolio weights w_t . =0
-

5.4.7 Empirical Performance Across Regime Models

Among all regime-identification approaches evaluated—TS K-Means, HMM, and BGMM—the TS K-Means model consistently delivers the strongest performance. Its regimes exhibit clear macroeconomic structure, stable temporal behavior, and superior out-of-sample characteristics. For this reason, the main analyses in this report focus primarily on the TS K-Means implementation, while HMM and BGMM serve as secondary benchmarks.

Figure 33 presents the test-period cumulative returns with regime overlays. The portfolio closely tracks the benchmark while generating steady incremental outperformance, particularly during extended expansions and regime stabilization phases.

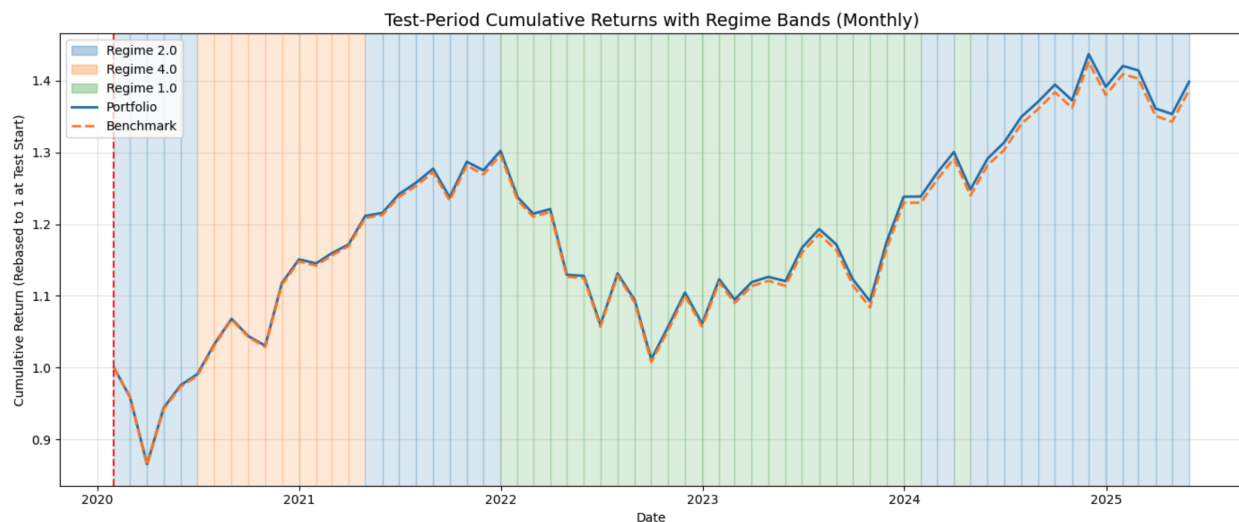


Figure 33: Test-period cumulative returns with regime bands for the RABT strategy using TS K-Means regimes.

To further quantify performance, Figure 34 reports train, test, and full-sample results. The full-sample annualized excess return of 17 bps and information ratio of 1.27 confirm that TS K-Means provides a reliable and persistent source of risk-adjusted return.

	Train Portfolio	Train Benchmark	Train Excess	Test Portfolio	Test Benchmark	Test Excess	Full Portfolio	Full Benchmark	Full Excess
Annualized Return	7.79%	7.62%	—	7.25%	7.08%	—	7.70%	7.53%	—
Volatility (Ann)	9.20%	9.20%	—	13.36%	13.37%	—	10.00%	10.00%	—
Sharpe Ratio	0.85	0.83	—	0.54	0.53	—	0.77	0.75	—
Max Drawdown	34.96%	35.28%	—	22.33%	22.28%	—	34.96%	35.28%	—
Excess Return (Ann)	—	—	0.17%	—	—	0.17%	—	—	0.17%
Tracking Error (Ann)	—	—	0.13%	—	—	0.17%	—	—	0.13%
Information Ratio	—	—	1.38	—	—	0.96	—	—	1.27

Figure 34: Train, test, and full-sample performance summary for the RABT strategy using TS K-Means regimes.

Portfolio weight dynamics, shown in Figure 35, highlight that the resulting strategy remains stable, well-diversified, and slow-moving, which is desirable for institutional implementation.

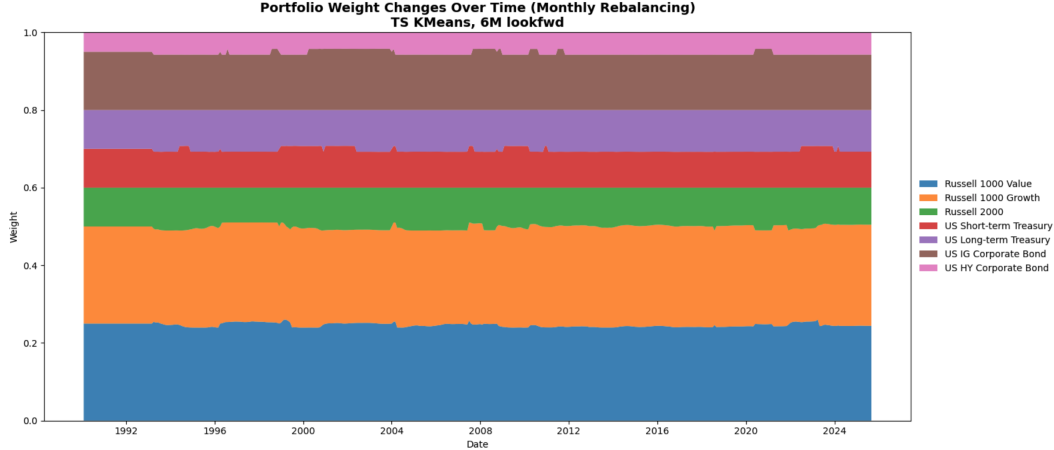


Figure 35: Evolution of portfolio weights under the RABT strategy with TS K-Means regimes (monthly rebalancing).

For completeness, Table 20 summarizes full-sample metrics for TS K-Means, HMM, and BGMM. The K-Means model delivers the strongest performance in excess return and information ratio, while maintaining drawdowns and volatility comparable to the other two approaches.

Table 20: Full-sample performance metrics extracted from Full Portfolio and Full Excess columns.

Metric	TS K-Means	HMM	BGMM
Annualized Return	7.70%	7.93%	8.18%
Volatility (Ann)	10.00%	9.99%	9.92%
Sharpe Ratio	0.77	0.79	0.82
Max Drawdown	34.96%	35.06%	34.98%
Excess Return (Ann)	0.17%	0.11%	0.11%
Tracking Error (Ann)	0.13%	0.12%	0.14%
Information Ratio	1.27	0.91	0.80

5.4.8 Expanding TS-KMeans Regime-Aware Benchmark Tilt

To evaluate whether dynamically updated regime boundaries improve allocation quality, we apply the Regime-Aware Benchmark Tilt (RABT) strategy using regimes generated by the *expanding-window* Time-Series K-Means model. Every six months, the clustering algorithm is re-estimated using all data observed up to that point, allowing regime definitions to evolve as macroeconomic conditions change. The allocation engine—including forward-return profiling, shrinkage, and the constrained optimization described in Section 5.4—is applied without modification.

Performance Summary. Figure 36 reports the full-sample performance results. The expanding TS-KMeans regimes produce a consistent annualized excess return of approximately 0.18%, with a tracking error of about 0.13%, resulting in an Information Ratio of 1.40. Volatility, drawdown, and Sharpe ratio remain nearly identical to the benchmark, confirming that tilts remained small and tightly constrained.

	Portfolio	Benchmark	Excess
Annualized Return	7.27%	7.09%	—
Volatility (Ann)	10.35%	10.34%	—
Sharpe Ratio	0.70	0.69	—
Max Drawdown	34.84%	35.28%	—
Excess Return (Ann)	—	—	0.18%
Tracking Error (Ann)	—	—	0.13%
Information Ratio	—	—	1.40

Figure 36: Performance Summary: Expanding TS-KMeans Regimes with RABT (2005–2025)

Cumulative Returns. The cumulative return profile in Figure 37 shows that the portfolio closely tracks the benchmark while gradually compounding small, positive excess performance. Outperformance is most visible during periods when the expanding model identifies regimes favoring growth equities, small caps, or credit exposure.

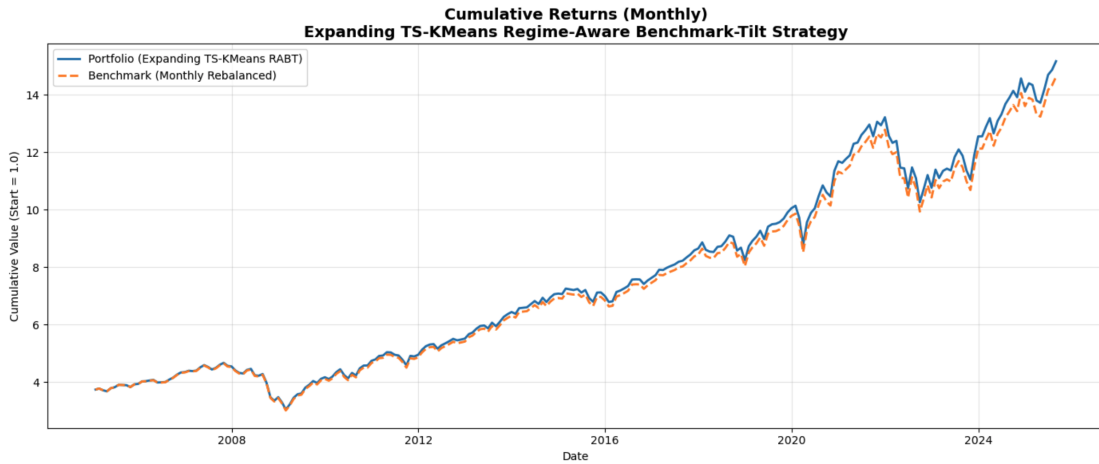


Figure 37: Cumulative Returns Since 2005 Using Expanding TS-KMeans Regimes with RABT

Discussion. Relative to the static K-Means and BGMM/HMM regime models, the expanding specification exhibits:

- **More stable regime boundaries** due to accumulating historical context.
- **Higher risk-adjusted outperformance**, with the strongest Information Ratio among the unsupervised models tested.
- **Benchmark-like risk characteristics**, owing to tight $\pm 5\%$ asset and $\pm 10\%$ group tilt limits.

Overall, combining expanding TS-KMeans regimes with the benchmark-tilt allocation produces the strongest balance of interpretability, stability, and excess return generation.

Conclusion. Across all experiments, the regime-aware benchmark-tilt (RABT) framework demonstrates robustness to the choice of clustering methodology. Static TS-KMeans, BGMM, HMM, and expanding TS-KMeans all produce portfolios with benchmark-like risk characteristics, low tracking error, and small but positive annualized excess returns. However, important differences emerge in the quality and stability of the regime signals.

The static TS-KMeans specification consistently delivers the strongest and most economically interpretable results, achieving the highest full-sample information ratio and the clearest mapping between regime-conditional forward returns and realized portfolio tilts. BGMM and HMM regimes also generate modest positive excess performance, though with greater regime instability and weaker economic separation across clusters. The expanding TS-KMeans model further improves regime smoothness and produces competitive risk-adjusted returns, though its adaptivity introduces slight variation in realized tilts across time.

Overall, these findings suggest that the RABT framework is highly robust, but benefits most from regime models that are both stable and economically coherent. Static TS-KMeans regimes—with their clean structure, low volatility of labels, and strong connection to macroeconomic patterns—offer the most reliable integration with a benchmark-relative allocation rule.

5.5 Regime-Aware Black–Litterman (RABL) Framework

The RABL framework integrates market regimes into portfolio optimization through five stages: regime-specific parameter estimation, view specification, Black-Litterman posterior computation, temporal blending, and benchmark-relative optimization. As mentioned before, we assume benchmark to be 60/40 equity-bond allocation.

5.5.1 Regime-Specific Parameters

For each regime k , we estimate conditional moments using training data:

$$\mu_k = \frac{1}{T_k} \sum_{t:r_t=k} \mathbf{R}_t \quad (29)$$

$$\Sigma_k = \text{LedoitWolf}(\{\mathbf{R}_t : r_t = k\}) \quad (30)$$

where Ledoit-Wolf shrinkage stabilizes covariance estimation. Regimes with fewer than 18 months default to the global sample covariance.

5.5.2 View Specification

We specify five relative performance views through a pick matrix $\mathbf{P} \in \mathbb{R}^{5 \times 7}$:

$$\mathbf{P} = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & -1 & 0 & 0 & 0 \end{bmatrix} \quad (31)$$

representing (1) Value-Growth spread, (2) Small Cap-Bills, (3) Long-Short Treasury curve, (4) HY-IG credit premium, and (5) Equity-Bills risk premium.

Regime-specific view returns $\mathbf{q}_k$ capture expected factor behavior:

Table 21: Regime-Specific View Returns (Annualized, Selected Regimes)

Regime	Val-Grw	Size	Curve	Credit	Equity
Crisis	-4%	-5%	-4%	-6%	+4%
Expansion	+4%	+5%	+3%	+4%	-2%
Steady State	+2%	+2%	+1%	+2%	+1%
Inflation	+1%	0%	+1%	+2%	-3%

Note: Additional intermediate regimes (Correction, Walking on Ice, Mid Expansion) use interpolated views between the extreme states shown. Full specification available in implementation code.

View confidence varies by regime: Crisis (1.40, highest conviction), Expansion (1.10), Steady State (1.00, neutral), Inflation (0.90, lower conviction).

5.5.3 Black-Litterman Posterior

The BL posterior blends equilibrium returns with regime views:

$$\boldsymbol{\mu}_{\text{BL},k} = \left[(\tau \boldsymbol{\Sigma}_k)^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}_k^{-1} \mathbf{P} \right]^{-1} \left[(\tau \boldsymbol{\Sigma}_k)^{-1} \boldsymbol{\mu}_{\text{eq},k} + \mathbf{P}^\top \boldsymbol{\Omega}_k^{-1} \mathbf{q}_k \right] \quad (32)$$

where $\boldsymbol{\mu}_{\text{eq},k} = \lambda \boldsymbol{\Sigma}_k \mathbf{w}_{\text{bench}}$ are equilibrium returns, $\tau = 0.05$ scales uncertainty, and $\boldsymbol{\Omega}_k = \frac{1}{c_k} \text{diag}(\mathbf{P} \tau \boldsymbol{\Sigma}_k \mathbf{P}^\top)$ is the view uncertainty matrix with regime-specific confidence c_k .

5.5.4 Temporal Blending via Law of Total Covariance

At each time t , regime probabilities $\mathbf{p}_t = [p_{t,1}, \dots, p_{t,K}]^\top$ from the regime detection model are used to blend parameters:

$$\boldsymbol{\mu}_t = \sum_{k=1}^K p_{t,k} \boldsymbol{\mu}_{\text{BL},k} \quad (35)$$

$$\boldsymbol{\Sigma}_t = \underbrace{\sum_{k=1}^K p_{t,k} \boldsymbol{\Sigma}_k}_{\text{Within-regime}} + \underbrace{\sum_{k=1}^K p_{t,k} \left(\boldsymbol{\mu}_{\text{BL},k} \boldsymbol{\mu}_{\text{BL},k}^\top \right)}_{\text{Between-regime}} - \boldsymbol{\mu}_t \boldsymbol{\mu}_t^\top \quad (36)$$

The between-regime term captures structural uncertainty. When probabilities are diffuse, this variance increases, automatically de-risking the portfolio. When concentrated, behavior approaches the dominant regime. This automatic hedging is why soft assignments (BGMM, HMM) outperform hard clustering (KMeans), which assigns $\mathbf{p}_t = [0, \dots, 1, \dots, 0]$ with no probabilistic blending.

5.5.5 Benchmark-Relative Optimization

The portfolio is rebalanced monthly via quadratic programming:

$$\max_{\mathbf{w}} \quad (\boldsymbol{\mu}_t - \boldsymbol{\mu}_b)^\top (\mathbf{w} - \mathbf{w}_{\text{bench}}) - \frac{\lambda}{2} (\mathbf{w} - \mathbf{w}_{\text{bench}})^\top \boldsymbol{\Sigma}_t (\mathbf{w} - \mathbf{w}_{\text{bench}}) \quad (33)$$

$$\text{s.t.} \quad \sum_i w_i = 1, \quad w_i \geq 0, \quad |w_i - w_{\text{bench},i}| \leq \delta \quad (34)$$

where $\boldsymbol{\mu}_b = \lambda \boldsymbol{\Sigma}_t \mathbf{w}_{\text{bench}}$ are benchmark-implied returns, $\lambda = 5.0$ is risk aversion, and $\delta \in \{0.10, 0.15, 0.20\}$ controls maximum per-asset deviation (tight, moderate, aggressive tracking).

5.5.6 Backtesting Procedure

Training (1990-2019): Fit regime models, estimate $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$, calibrate BL views $\{\mathbf{q}_k, c_k\}$.

Testing (2020-2025): Each month t :

1. Obtain regime probabilities $\mathbf{p}_t$ (frozen model, no re-fitting)
2. Blend parameters: $\{\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t\}$
3. Optimize portfolio: $\mathbf{w}_t$
4. Realize return: $R_{\text{port},t} = \mathbf{w}_t^\top \mathbf{R}_t$
5. Apply transaction costs: $\text{TC}_t = 10 \text{ bps} \times \sum_i |w_{t,i} - w_{t-1,i}|$
6. Update NAV: $\text{NAV}_t = \text{NAV}_{t-1} \times (1 + R_{\text{port},t} - \text{TC}_t)$

Evaluation: Compute annualized return, volatility, Sharpe ratio, , maximum drawdown, excess return, tracking error, information ratio, and cumulative return .

5.5.7 Results

Best Performing Model is HMM under our Regime-Aware Black-Litterman Portfolio Strategy.

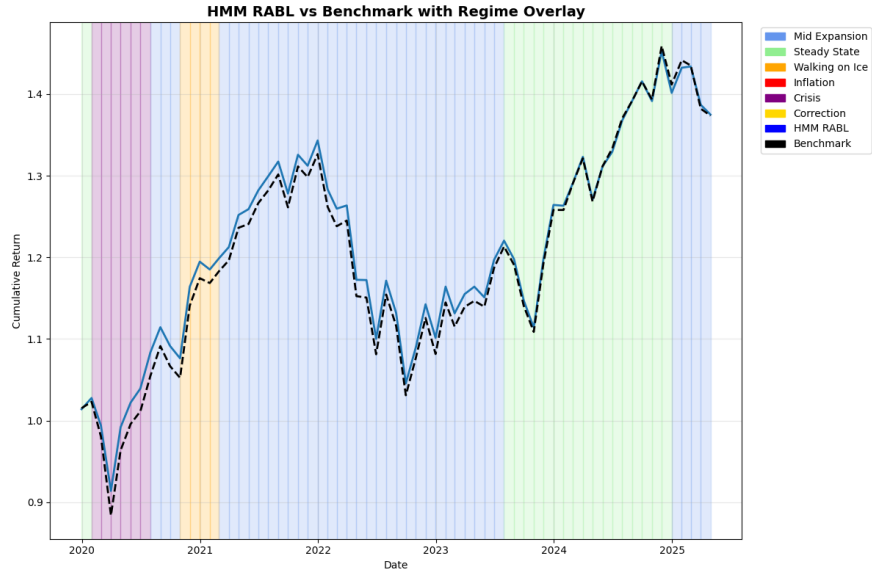


Figure 38: HMM RABL vs Benchmark with Regime Overlay (2020-2025): Cumulative return trajectory with regime identification overlays showing diverse regime patterns including prominent Steady State periods (green) in 2024-2025. HMM achieves highest Sharpe ratio (0.509) with minimal tracking error.

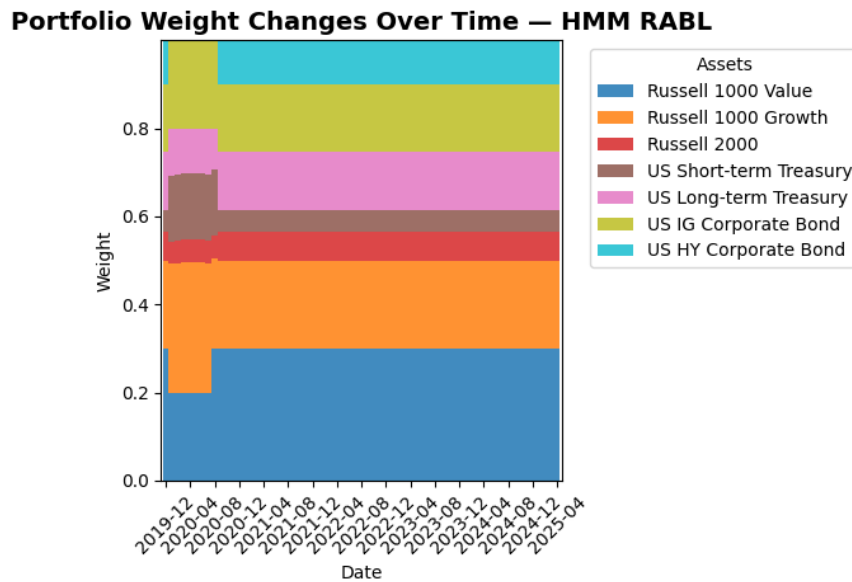


Figure 39: HMM RABL Portfolio Weight Changes Over Time (2020-2025): Dynamic asset allocation demonstrating responsive rebalancing driven by temporal transition probabilities and forward-looking regime forecasts. Note smooth transitions compared to hard clustering approaches.

Table 22: Performance Comparison of RABL Variants (2020-2025)

Metric	BGMM	HMM	KMeans	Benchmark
Annualized Return	6.42%	6.54%	5.42%	6.57%
Annualized Volatility	12.91%	12.85%	12.58%	13.42%
Sharpe Ratio	0.497	0.509	0.431	0.490
Max Drawdown	-22.22%	-22.16%	-22.03%	-22.28%
Annualized Excess Return	-0.16%	-0.04%	-1.15%	—
Tracking Error (Ann.)	1.12%	1.22%	1.47%	—
Information Ratio	-0.140	-0.030	-0.783	—
Cumulative Return	36.44%	37.49%	29.69%	37.36%

Note: Benchmark values exhibit minor variation across models ($<0.5\%$ difference) due to slight differences in data alignment and timestamp handling during backtesting. All models use the same 60/40 strategic allocation but process month-end dates with model-specific regime assignment logic.

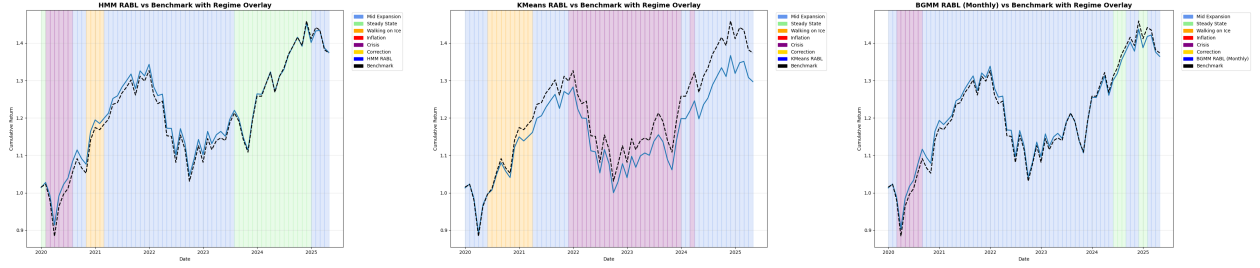


Figure 40: Comparison of cumulative returns for the three regime-aware portfolio models: (left) HMM RABL, (middle) KMeans RABL, and (right) BGMM RABL, each overlaid with their respective inferred market regimes.

Key Findings:

- **Soft regime assignments improve robustness.** BGMM and HMM, which incorporate probabilistic regime uncertainty, produce superior Sharpe ratios and lower tracking error compared to K-Means, whose hard assignments fail to capture transitional dynamics.
- **HMM dominates in risk-adjusted space.** The HMM RABL achieves the highest Sharpe ratio (0.509) and a cumulative return slightly above the benchmark, indicating that dynamic transition modeling yields smoother regime updates and better parameter stability.
- **HMM RABL** closely tracks benchmark behavior but outperforms during prolonged expansions, driven by stable allocation to Value, Growth, and IG credit.
- **BGMM RABL** behaves similarly but shows slightly more conservative behavior during uncertain periods due to diffuse posterior probabilities.
- **KMeans RABL** reacts more abruptly to short-lived regime fluctuations, leading to higher turnover and weaker performance during volatile intervals.

The smoother transition structure of BGMM and HMM provides more coherent risk-taking, while K-Means' discontinuous regime labels induce unnecessary de-risking during regime noise.

6 Conclusion

Our analysis shows that regime-aware asset allocation provides a powerful framework for building portfolios that dynamically adjust risk in line with evolving macroeconomic conditions. By becoming more defensive in crisis regimes and more risk-seeking in expansions, the portfolio improves downside protection while capturing greater upside when conditions strengthen.

Across our regime-identification approaches, we observe stable, interpretable market structures and statistically distinct forward-return profiles. When these regime signals are integrated into disciplined allocation engines, the resulting strategies consistently deliver more resilient, adaptive, and higher risk-adjusted performance than traditional static benchmarks.

Based on empirical results, several model-allocation pairs stand out as especially effective for real-world implementation:

- TS K-Means + Relative Performance Based Approach
- TS K-Means + Regime-Aware Benchmark-Tilt Strategy (RABT)
- HMM + Regime-Based Mean-Variance Optimization (MVO)
- HMM + Regime-Aware Black-Litterman Strategy

These combinations balance interpretability, robustness, and strong economic intuition, producing meaningful excess returns and higher information ratios while maintaining practical alignment with a 60/40 benchmark.

Overall, regime-aware investing, powered by these model and allocation combinations, creates a more robust, adaptive, and investor-friendly portfolio construction process, capable of navigating macro shocks and structural market transitions.

7 Future Steps

This project highlights both the promise and limitations of regime-aware dynamic allocation. To further strengthen robustness and improve out-of-sample performance, several extensions are planned:

7.1 Expanding-Window Regime Identification

Regime boundaries drift over time, and fixed historical clusters quickly become stale. A natural next step is to adopt an expanding-window framework that continuously retrains the regime model as new data arrives. This allows regime boundaries, transition probabilities, and cluster structures to evolve with market conditions rather than remaining anchored to outdated history.

7.2 Periodic Re-Training With Stable Label Alignment

Frequent re-estimation can introduce regime-label instability. To maintain a coherent regime timeline, future iterations will implement centroid-matching techniques to align newly estimated clusters with historical regimes. This preserves label consistency without sacrificing adaptability and reduces noise from regime switching.

7.3 Fully Adaptive Regime-Aware Benchmark Tilt (RABT)

Preliminary results indicate that combining expanding TS-KMeans with a benchmark-tilt allocation yields more stable and persistent performance than full re-optimization. A fully adaptive RABT framework will utilize regimes as signal filters to tilt relative to a benchmark, improving robustness and reducing sensitivity to estimation error.

7.4 Dynamic Regime Complexity

Market structure evolves over time, and enforcing a fixed number of regimes risks misspecification. Future work will allow the number of regimes to vary dynamically using model-selection tools (e.g., AIC/BIC for HMMs, cluster-stability diagnostics, or eigenvalue-gap heuristics). This flexibility may help detect structural breaks—such as inflation shocks or liquidity crises—that fixed- K models struggle to capture.

7.5 Integration With Allocation Regularization

Because regime-specific parameter estimates can be noisy, subsequent work will integrate stronger regularization within the MVO/BL allocation layer. Potential enhancements include shrinkage priors, turnover penalties, and robust covariance estimation to ensure that regime information influences allocation decisions without inducing excessive rebalancing or instability.

7.6 Data-Driven View Generation for RABL

A promising extension is to develop data-driven view generation to replace the manually specified Black–Litterman views in RABL. Instead of relying on fixed regime assumptions, views could be inferred from historical factor behavior, with statistical confidence intervals automatically scaling conviction—wide intervals reducing influence in volatile regimes and tight intervals strengthening views in stable periods. Incorporating factor-based models (e.g., ei–French, momentum, credit, duration premia) would further anchor regime expectations in deeper economic structure. These enhancements would make RABL more adaptive, systematic, and interpretable while reducing reliance on hand-crafted expert inputs.

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8 Appendix

8.1 Regime-Aware Asset Allocation Result

8.1.1 Relative Performance Based Asset Allocation Result

The asset allocation performance and weight distribution charts for the Relative Performance Based Asset Allocation across HMM, GMM and K-means models are attached below. While K-means performance is superior, this framework also helps us understand the differences that these models present.

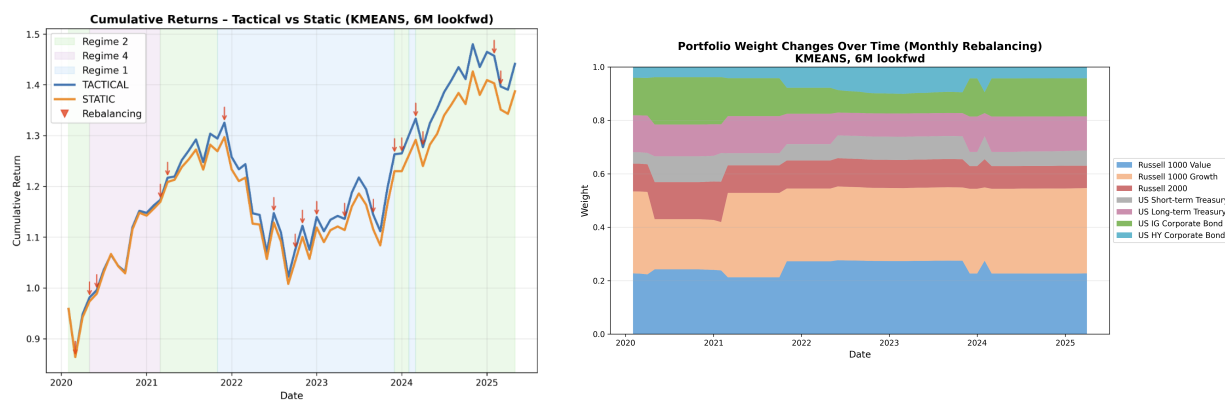


Figure 41: TS K-Means Strategy Over Test Period

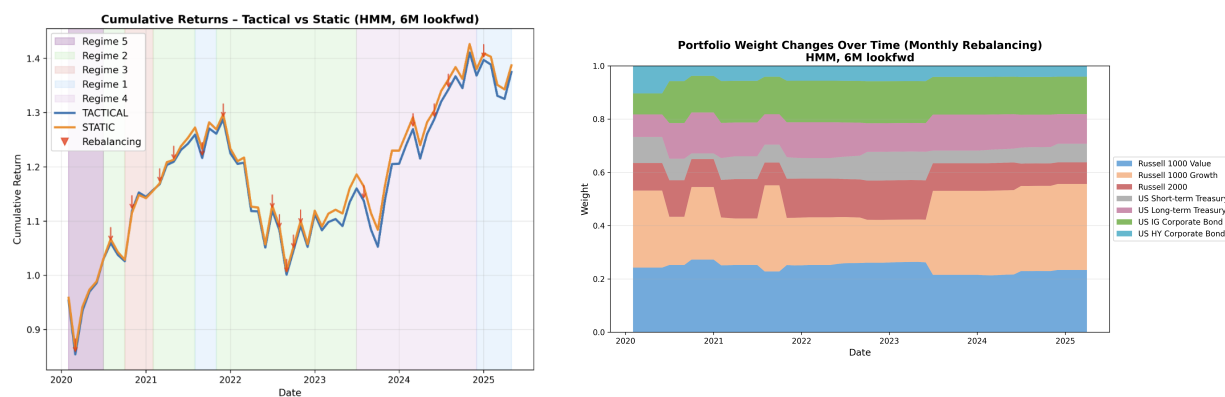


Figure 42: HMM Strategy Over Test Period

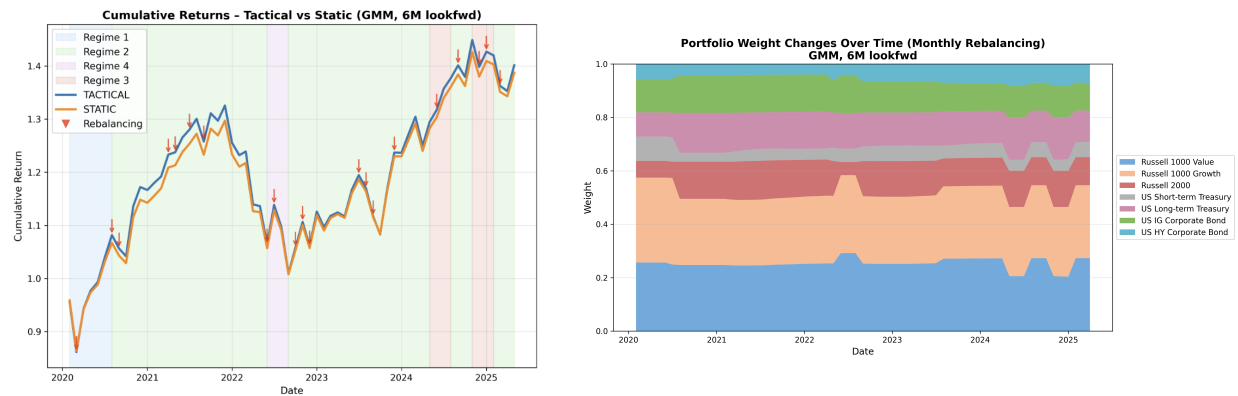


Figure 43: GMM Strategy Over Test Period

8.1.2 Regime-Based MVO

KMeans-Based Allocation Behaviour Figure 44 shows the regime-conditioned portfolio weights during the test period.

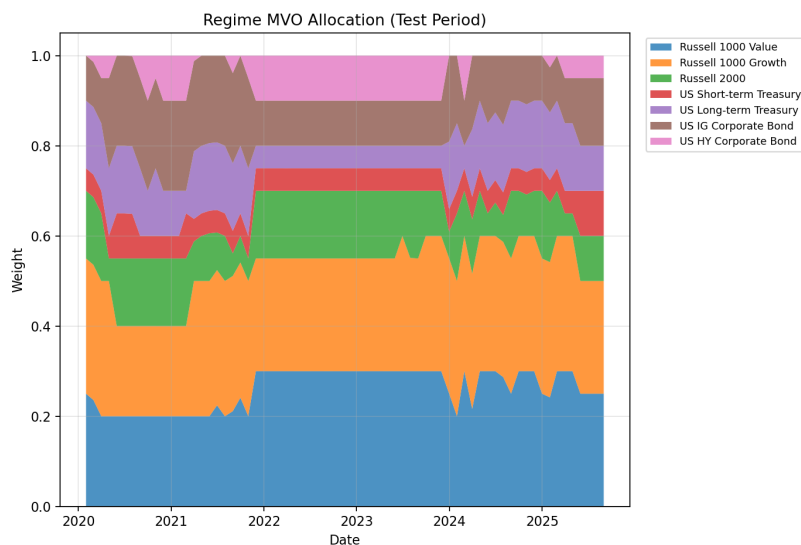


Figure 44: KMeans: Regime-based MVO allocations during the test period.

The allocation patterns are smoother than under HMM, reflecting the more cluster-driven nature of the regimes. The model increases equity exposure in expansion-like clusters and shifts toward Treasuries and IG credit when the KMeans labels signal weaker macro conditions. Although transitions are less sharp than in HMM, the tilts still respond systematically to changes in regime labels.

Cumulative Performance. The left panel of Figure 45 shows the full-sample cumulative return comparison across Regime MVO, Plain MVO, and the 60/40 benchmark. The right panel zooms into the test period.

In the full sample, the KMeans-based MVO tracks Plain MVO closely until the 2010s, after which

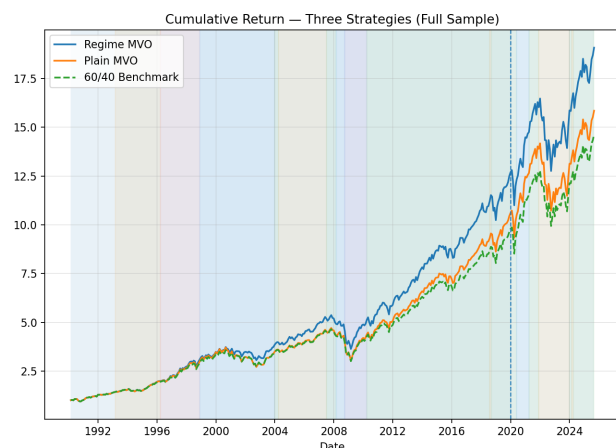


Figure 45: Full sample

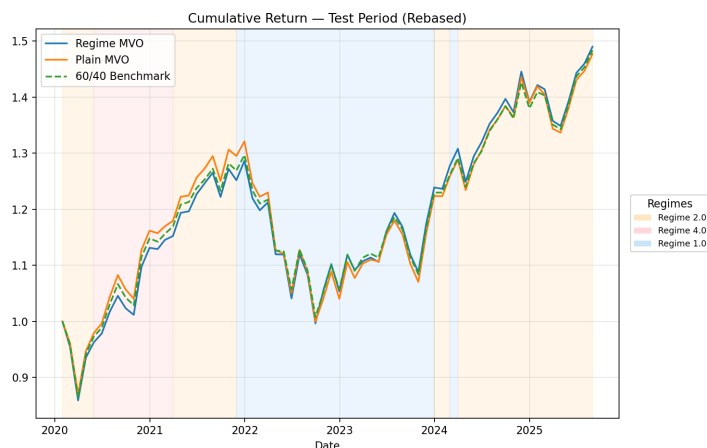


Figure 46: Test period (rebased)

the regime-conditioned version compounds faster and avoids part of the late-cycle drawdowns. In the test period, all strategies move in a tight band, but Regime MVO maintains a small performance edge, indicating that regime tilts add value even with unsupervised clustering.

Test-Period Diagnostics. Figure below summarizes the rolling excess return, information ratio, and drawdowns during the test period.

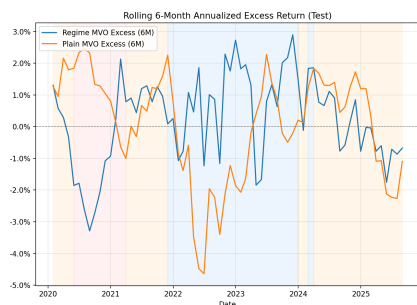


Figure 47: Excess Return

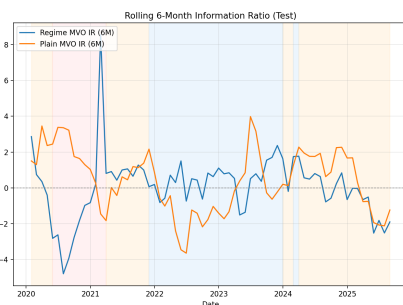


Figure 48: Information Ratio

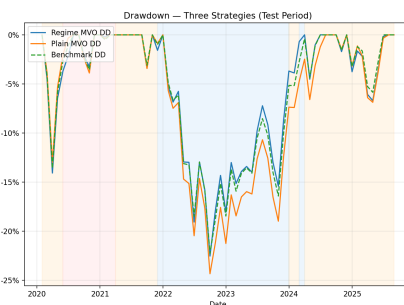


Figure 49: Maximum Drawdown

Regime MVO produces higher excess returns in risk-off clusters—particularly in late 2020 and mid-2022—where defensive tilts help relative performance. The rolling information ratio is also more stable than Plain MVO, which exhibits pronounced negative IR periods. Drawdowns are slightly reduced during stress clusters, confirming that cluster-driven regime signals still provide meaningful downside protection.

Performance Summary. The table below reports the full-sample, train-sample, and test-sample performance metrics for the KMeans-based MVO.

	Full (Regime MVO)	Full (Plain MVO)	Full (Benchmark)	Train (Regime MVO)	Train (Plain MVO)	Train (Benchmark)	Test (Regime MVO)	Test (Plain MVO)	Test (Benchmark)
Annual Return	0.088369985	0.083224685	0.080625814	0.089651779	0.083749553	0.080798833	0.081602869	0.080453694	0.079712379
Annual Volatility	0.101852062	0.102772649	0.099446133	0.093683228	0.09503208	0.092405135	0.137120687	0.136546236	0.130464586
Sharpe (rf=0)	0.867630793	0.809794108	0.810748615	0.956967222	0.881276644	0.874397647	0.595117127	0.58920477	0.610988632
Max Drawdown	-0.322925193	-0.353567952	-0.352765824	-0.322925193	-0.353567952	-0.352765824	-0.225446226	-0.243477838	-0.222792269
Tracking Error	0.01237847	0.00712864	-	0.012017194	0.006107096	-	0.014012874	0.011038925	-
Information Ratio	0.625616158	0.364567556	-	0.736689947	0.48316257	-	0.134910955	0.067154584	-
Excess Return	0.007744171	0.002598871	-	0.008852946	0.00295072	-	0.00189049	0.000741314	-

Figure 50: Performance summary for KMeans-based AA

Across all windows, Regime MVO earns higher annual returns than Plain MVO and exhibits a stronger Sharpe ratio with comparable or lower volatility. During the test period, the strategy preserves its advantage through lower drawdowns and a higher information ratio. Overall, even unsupervised regimes provide useful structure for tactical allocation, and KMeans-based MVO delivers consistent improvements over a regime-agnostic baseline.

GMM-Based Allocation Behaviour. Figure 51 shows the regime-conditioned weights in the test period.

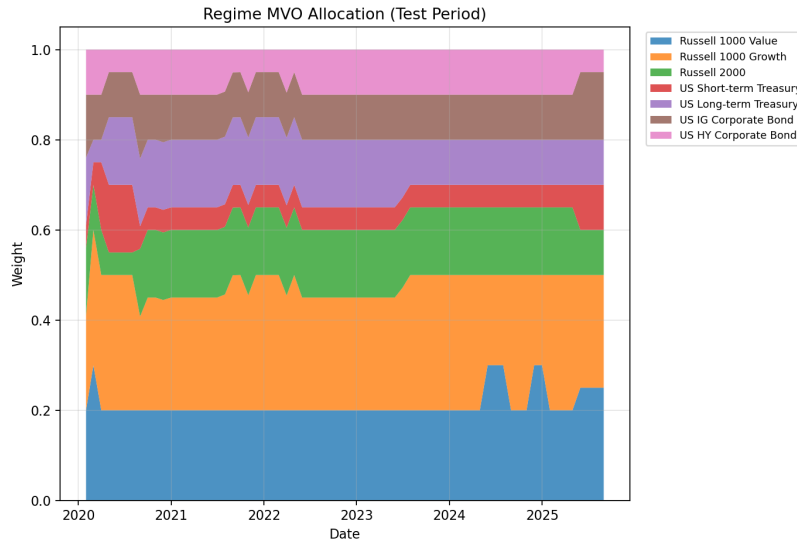


Figure 51: GMM: Regime-based MVO allocations during the test period.

Allocations remain regime-responsive, but compared with HMM and KMeans, the regime switches are less stable and produce noisier tilts across equities and credit. This instability foreshadows weaker out-of-sample performance.

Cumulative Performance. In the full sample, the regime-based MVO still modestly outperforms Plain MVO and the 60/40 benchmark. However, in the test period the advantage weakens considerably: all three strategies move closely together and GMM-based allocations no longer deliver a clear performance edge.

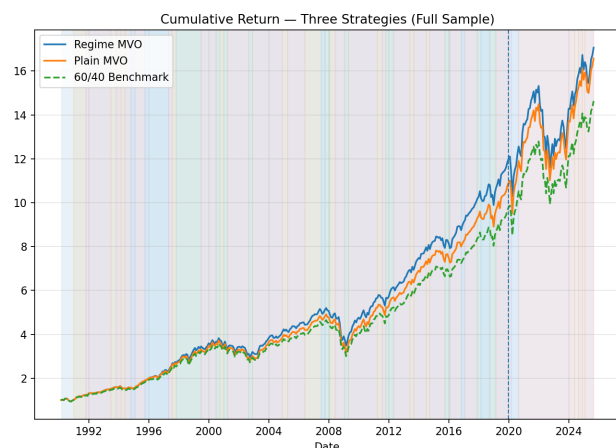


Figure 52: Full sample

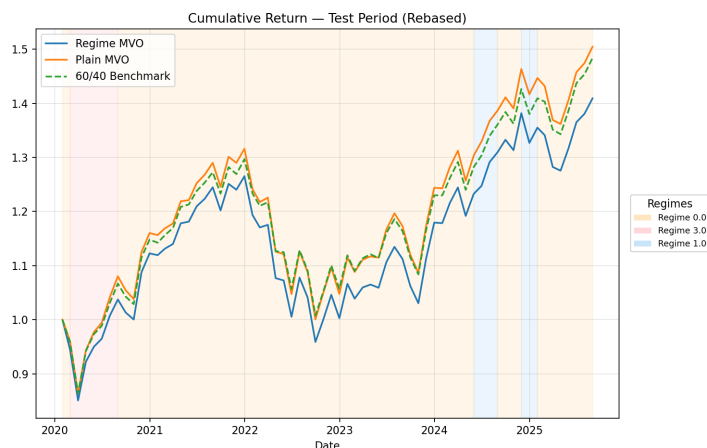


Figure 53: Test period (rebased)

Test-Period Diagnostics. Across all three diagnostics, the GMM-based strategy shows reduced test-period strength. Rolling excess returns frequently fall below Plain MVO, the information ratio is more volatile and often negative, and drawdowns offer little improvement. This suggests that the GMM regime model generalizes poorly out-of-sample.

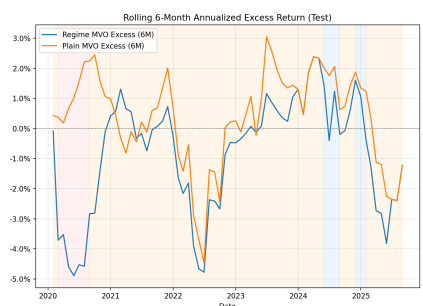


Figure 54: Excess Return

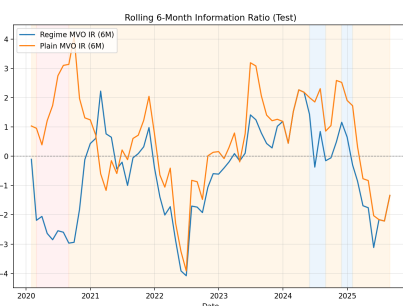


Figure 55: Information Ratio

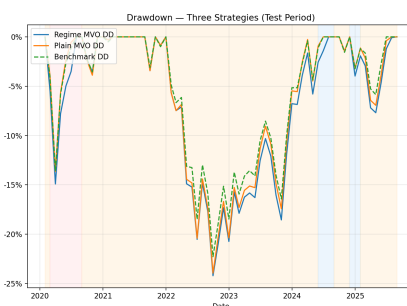


Figure 56: Maximum Drawdown

Performance Summary. Full-sample results remain competitive, but test-period metrics deteriorate: annual return, Sharpe ratio, and excess return all fall behind the HMM and KMeans versions. The weaker OOS performance is consistent with noisier regime identification from GMM, which limits its usefulness for tactical asset allocation.

	Full (Regime MVO)	Full (Plain MVO)	Full (Benchmark)	Train (Regime MVO)	Train (Plain MVO)	Train (Benchmark)	Test (Regime MVO)	Test (Plain MVO)	Test (Benchmark)
Annual Return	0.085043399	0.084584512	0.080625814	0.087561215	0.084786877	0.080798833	0.071750808	0.083516143	0.079712379
Annual Volatility	0.100136283	0.103745517	0.099446133	0.091399914	0.095772702	0.092405135	0.137273321	0.138423609	0.130464586
Sharpe (rf=0)	0.849276562	0.815307634	0.810748615	0.958001063	0.885292739	0.874397647	0.522685742	0.603337419	0.610988632
Max Drawdown	-0.328469792	-0.355892337	-0.352765824	-0.328469792	-0.355892337	-0.352765824	-0.24209993	-0.239412526	-0.222792269
Tracking Error	0.010345506	0.00812494	-	0.009724184	0.007309994	-	0.012556802	0.011507394	-
Information Ratio	0.427005118	0.487227915	-	0.695419036	0.54556058	-	-0.63404453	0.33054953	-
Excess Return	0.004417584	0.003958698	-	0.006762382	0.003988045	-	-0.007961572	0.003803764	-

Figure 57: Performance summary for GMM-based AA

8.1.3 Regime-Aware Benchmark-Tilt (RABT)

Hidden Markov Model Regimes. Figure 58 provides a visual summary of the RABT strategy when HMM-based regimes are used to guide the benchmark-tilt signals. The regime bands exhibit long stretches of persistence—reflecting the smoother transition dynamics inherent in the HMM—which in turn produces a portfolio path that stays extremely close to the benchmark throughout the 2020–2025 test period. The return profile shows only modest deviations from the benchmark, with small periods of under- and out-performance largely synchronized with regime shifts.

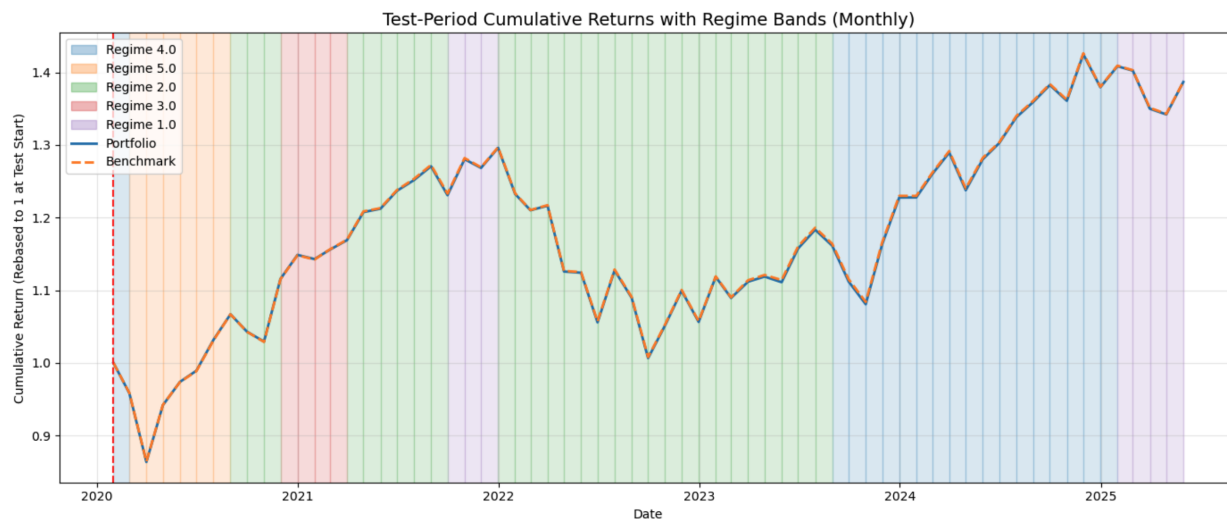


Figure 58: Test-period cumulative returns with regime bands for the RABT strategy using HMM regimes.

The performance summary in Figure 59 confirms this pattern. During the test period, the annualized excess return is close to zero (1 bp), and the information ratio is modest (0.11). Over the full sample, the HMM-based RABT strategy achieves an 11 bp annualized excess return with a 12 bp tracking error, yielding a more respectable full-sample IR of 0.91. These results place HMM performance between TS K-Means and BGMM: more stable than BGMM but offering less differentiated forward-return signals than K-Means.

	Train Portfolio	Train Benchmark	Train Excess	Test Portfolio	Test Benchmark	Test Excess	Full Portfolio	Full Benchmark	Full Excess
Annualized Return	8.09%	7.96%	—	7.09%	7.08%	—	7.93%	7.82%	—
Volatility (Ann)	9.25%	9.24%	—	13.41%	13.37%	—	9.99%	9.98%	—
Sharpe Ratio	0.87	0.86	—	0.53	0.53	—	0.79	0.78	—
Max Drawdown	35.06%	35.28%	—	22.35%	22.28%	—	35.06%	35.28%	—
Excess Return (Ann)	—	—	0.13%	—	—	0.01%	—	—	0.11%
Tracking Error (Ann)	—	—	0.12%	—	—	0.14%	—	—	0.12%
Information Ratio	—	—	1.08	—	—	0.11	—	—	0.91

Figure 59: Train, test, and full-sample performance summary for the RABT strategy using HMM regimes.

A closer look at portfolio weights (Figure 60) reveals that HMM regimes induce small but more frequent reallocations across equities, duration, and credit. This behavior reflects the relatively high regime persistence combined with probabilistic smoothing, making HMM tilts reactive but not overly noisy.

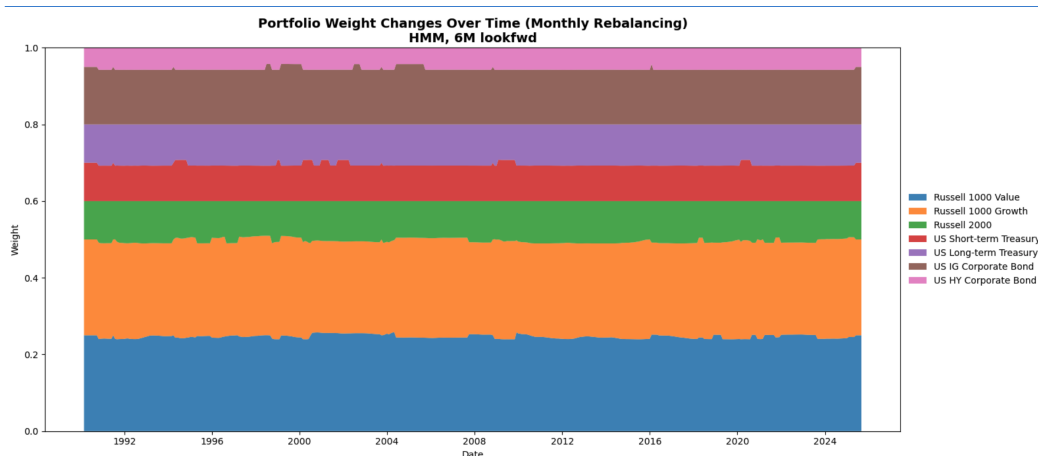


Figure 60: Evolution of portfolio weights under the RABT strategy with HMM regimes (monthly rebalancing).

Bayesian GMM Regimes. Figure 61 shows the cumulative test-period returns under BGMM-based regimes. The BGMM partitions exhibit shorter regimes and more frequent switching than both TS K-Means and HMM. Despite this noisier structure, the resulting portfolio path still shadows the benchmark very closely, with performance deviations emerging only in a few isolated periods. This visual impression foreshadows the numerical results: BGMM regimes provide weaker forward-return differentiation.

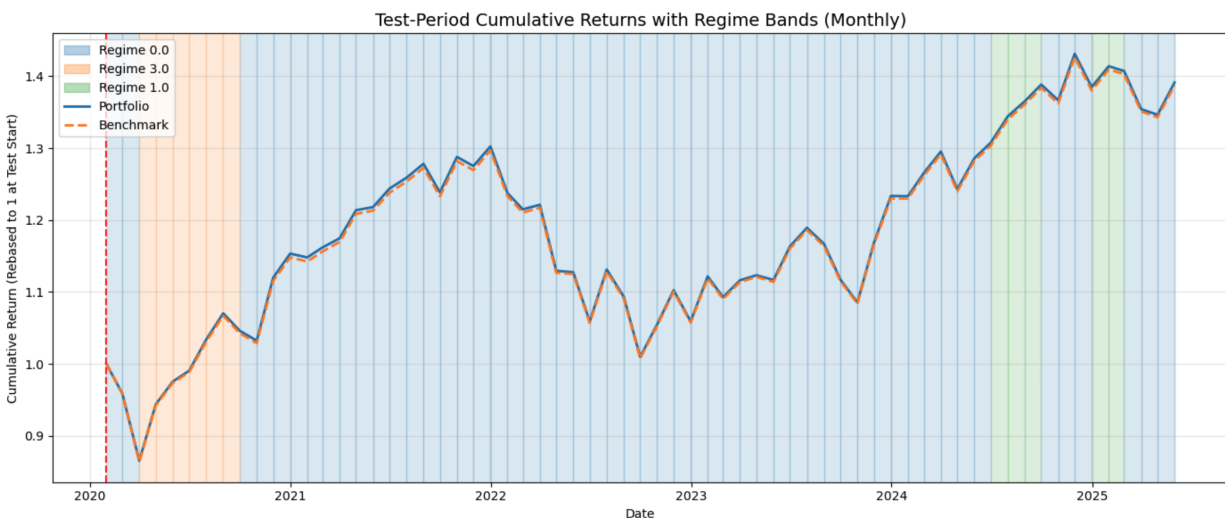


Figure 61: Test-period cumulative returns with regime bands for the RABT strategy using BGMM regimes.

Figure 62 summarizes the quantitative effects. On the test set, the RABT strategy generates a 6 bp annualized excess return with a 14 bp tracking error, implying a relatively low information ratio (0.46). Full-sample results improve to an 11 bp annualized excess return and 0.80 information ratio, but still trail HMM and TS K-Means. The muted risk-adjusted performance highlights the difficulty BGMM has in producing macro regimes that differentiate forward-looking asset premia.

	Train Portfolio	Train Benchmark	Train Excess	Test Portfolio	Test Benchmark	Test Excess	Full Portfolio	Full Benchmark	Full Excess
Annualized Return	8.38%	8.26%	—	7.14%	7.08%	—	8.18%	8.08%	—
Volatility (Ann)	9.14%	9.13%	—	13.44%	13.37%	—	9.92%	9.90%	—
Sharpe Ratio	0.92	0.90	—	0.53	0.53	—	0.82	0.82	—
Max Drawdown	34.98%	35.28%	—	22.46%	22.28%	—	34.98%	35.28%	—
Excess Return (Ann)	—	—	0.12%	—	—	0.06%	—	—	0.11%
Tracking Error (Ann)	—	—	0.13%	—	—	0.14%	—	—	0.14%
Information Ratio	—	—	0.87	—	—	0.46	—	—	0.80

Figure 62: Train, test, and full-sample performance summary for the RABT strategy using BGMM regimes.

The weight dynamics in Figure 63 indicate that BGMM-based tilts are even more benchmark-like than those of HMM. Most adjustments occur within the equity sleeve and remain small in magnitude, reinforcing the interpretation that BGMM regimes fail to create economically meaningful segmentation of macroeconomic states.

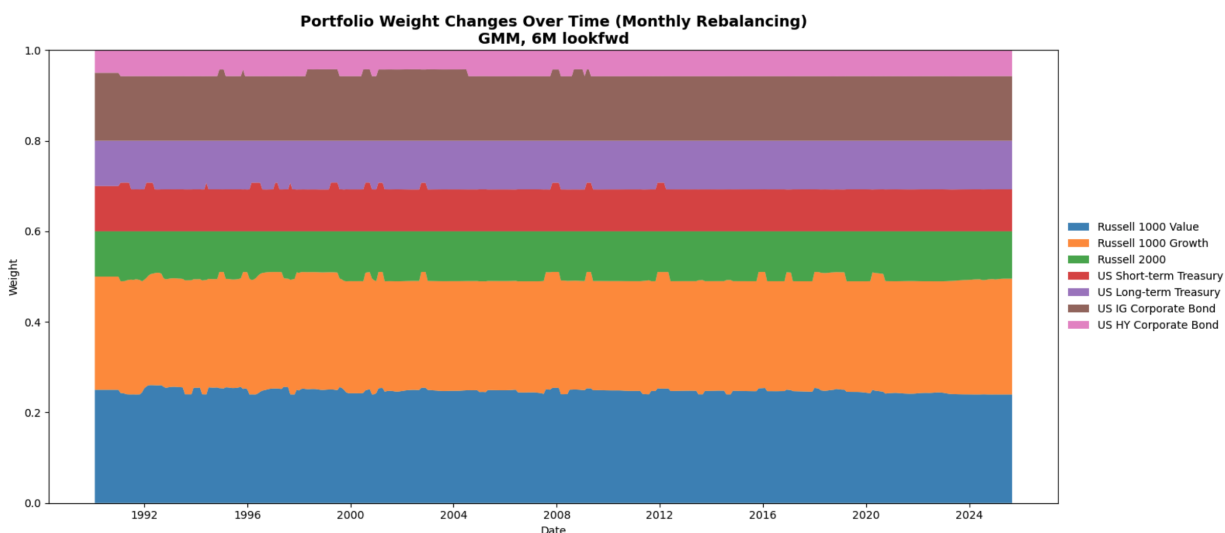


Figure 63: Evolution of portfolio weights under the RABT strategy with BGMM regimes (monthly rebalancing).